

AP MACROECONOMICS

MASTERY GUIDE v3

*Built around 17 actual practice questions.
Pattern recognition. Calculation fluency. Chain reasoning.*

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Goal: 5

The Battle Plan — Hour by Hour

This guide is structured for 5–7 hours total study across 2 days. Built around the 17 actual practice questions you uploaded — every pattern that showed up there is drilled here, in the order you'll encounter the material.

Day 1 — 3 to 4 hours

0:00–0:30 — Foundations + Connected Macro Model (Part 1)

Read it slowly. The hub diagram is your safety net for every chain question.

By the end: you can name every market, every axis, and how each feeds the next.

0:30–1:15 — Question Pattern Library (Part 2)

The most important new section. Every MCQ archetype on the AP exam, with the underlying pattern named.

Worked answers to all 15 of the actual MCQs you got. By the end: you'll see the trap before you read the question.

If pressed for time, you can pause this around 1:00 and skip to Part 3 — but come back to it on Day 2.

1:15–1:45 — Calculation Fluency Block (Part 3)

Every calculation the exam tests: GDP, real GDP, deflator, CPI, inflation, U rate, LFPR, multipliers, money creation, real interest rate, quantity theory, per capita.

Drill each until you can do them in <30 seconds.

1:45–2:45 — Unit 3 Deep + Unit 4 Deep (Parts 5 & 6, 60 min)

Unit 3 (AD/AS hub). Unit 4 (money + two interest rates — your weak point).

Critical distinction in Unit 4: money market = nominal rate, Fed-driven. Loanable funds = real rate, saver/borrower-driven. Half of Unit 4 traps target this confusion.

2:45–3:30 — Unit 5 Deep (Part 7, 45 min)

Highest weighted unit (20–30%) AND your weak point. Phillips curve, crowding out, money growth, growth.

Unit 5 is where Units 3 and 4 stop being separate — it's their synthesis.

3:30–4:00 — Compressed Units 1, 2, 6 (Part 8)

Lower-weight, mostly mechanical. Skim if foundations feel solid; read carefully if any one looks rusty.

End of Day 1: every unit covered, depth on the high-weight ones.

Day 2 — 2 to 3 hours

0:00–0:20 — Indeterminacy Mastery + Graph Reading Drills (Parts 4 & 9)

The 'both curves shift' rule applied across all six markets. Q9 in your practice (both contractionary) tests exactly this.

Graph reading drills mirror Q1 (PPC) and Q14 (Phillips) — the exam loves these.

0:20–0:50 — Master Chain Library (Part 10)

12 named cascades. Every 'trace through all markets' FRQ maps to one of these.

Memorize. By the end: you can recite each chain blind.

0:50–1:50 — Scenario Drills (Part 11, 30 scenarios)

Cover answer first. Reason chain. Check.

If you miss any reasoning step, flag it and re-read the relevant section.

This is where mastery is built — under simulated FRQ time pressure.

1:50–2:50 — FRQ Mastery + Timed Practice (Parts 12 & 13)

Part 12 includes worked solutions to BOTH actual FRQs from your practice (Fehran + Zeetoland).

Part 13: 30 MCQs in 35 min + 1 long FRQ in 25 min. Timed.

Last 10 min: glance at Part 14 (one-page cheat sheet).

Exam Morning — 10 minutes (optional)

Final review

Look at Part 14 (One-Page Cheat Sheet) ONLY. Don't try to learn anything new.

Better to skip this than to panic-cram. Trust the work.

Color key

How to read this guide

Cream box = a formula you must know cold.

Soft blue = a worked example. Solve it first, then check.

Soft red = a trap that costs students points every year.

Soft green = a strategy/time tactic.

Soft purple = a memory aid.

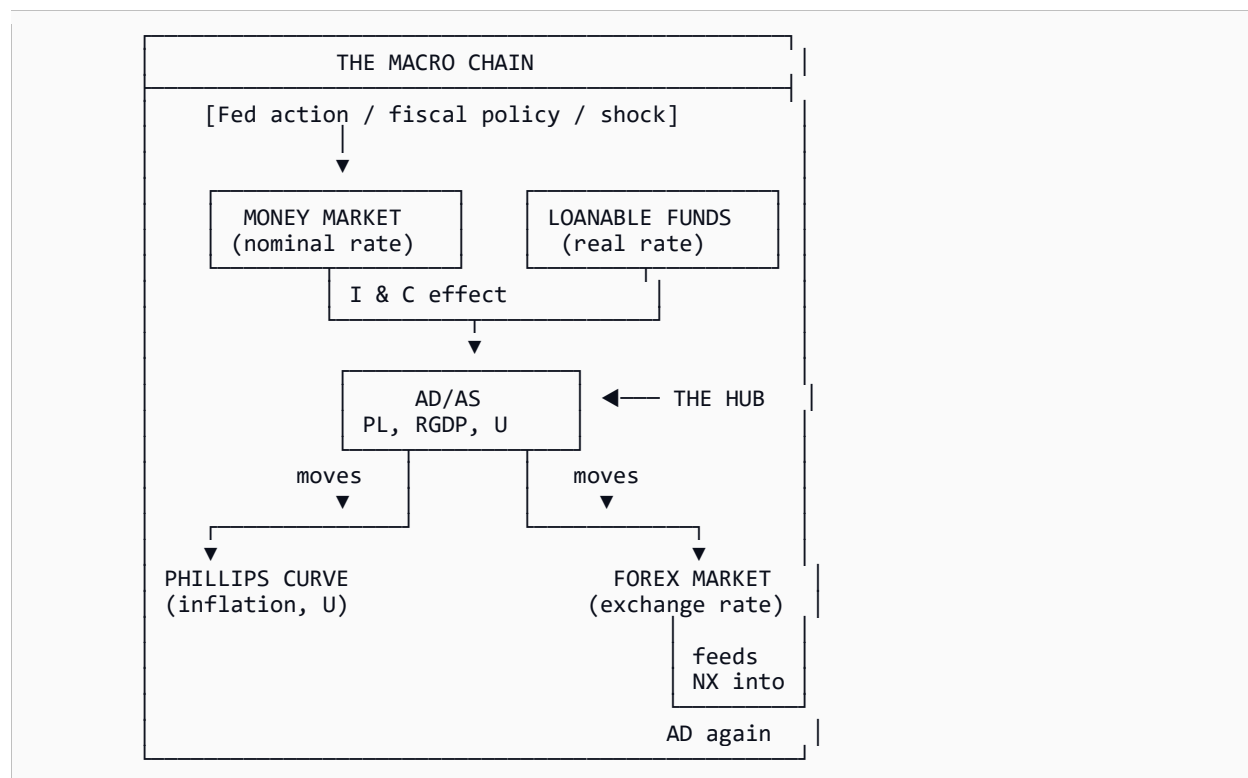
Amber = a key insight that ties multiple ideas together. Most important boxes.

Part 1 — The Connected Macro Model

Most students study macro as 6 separate units. Students who score 5s study it as ONE connected model. This section is the wiring diagram. Every other section refers back to it.

1.1 — The Six Markets, One Map

The AP Macro exam has six graphs. They are NOT independent. Shocks cascade through them in a fixed sequence:



Why this map matters

On every 'trace through all markets' FRQ, you do exactly this:

1. Identify the initial shock and which market it hits FIRST.
2. Read the equilibrium change in that market.
3. Follow the arrow to the next market.
4. Repeat until you reach AD/AS, then read off PL, RGDP, U.
5. Trace the AD/AS result into Phillips curve and (if needed) forex.

Different shocks enter at different points: monetary policy enters at Money Market. Fiscal enters at Loanable Funds. Confidence shocks enter directly at AD. The chain after the entry point is the same.

1.2 — The Six Graphs at a Glance

Graph	Y-axis	X-axis	What shifts the upward curve	What shifts the downward curve
Supply & Demand	Price	Quantity	Input prices, productivity, sellers, taxes, expectations	TRIBE: Tastes, Related goods, Income, Buyers, Expectations
AD/AS	Price Level	Real GDP	SRAS: input prices, productivity, expected inflation. LRAS: capital, labor, tech, human capital	AD: C, I, G, X-M shifters
Money Market	Nominal int. rate	Q of money	MS: Fed buys/sells bonds, RR, discount rate, IORB	MD: real GDP, price level
Loanable Funds	Real int. rate	Q of LF	SLF: savings, capital inflow, gov surplus	DLF: gov deficit, business confidence, investment incentives
Phillips Curve	Inflation rate	Unemployment	LRPC at NRU. SRPC shifts: expected inflation, supply shocks	Movement along SRPC: AD shifts only
Forex Market	Exchange rate	Q of currency	S_USD: US income, US inflation, US tastes, US rates ↓	D_USD: foreign income, foreign tastes, US rates ↑

1.3 — Foundations Compressed

The four shift outcomes (every market)

Burn this in — it solves Q2 and many others

D right: ↑P, ↑Q.

D left: ↓P, ↓Q.

S right: ↓P, ↑Q.

S left: ↑P, ↓Q.

BOTH SHIFTS, SAME DIRECTION (both right or both left): Q determined, P INDETERMINATE.

BOTH SHIFTS, OPPOSITE DIRECTIONS: P determined, Q INDETERMINATE.

This rule applies to EVERY market on the exam — goods, money, loanable funds, forex, reserves.

One rule, six applications.

The eight foundational facts

- **Comparative advantage = LOWER opportunity cost.** OUTPUT table: OC = Other/Own. INPUT table: OC = Own/Other. Trade benefits both when terms fall BETWEEN the two countries' OCs.

- **PPC:** on curve = efficient. Inside = recession/inefficiency. Outside = unattainable. Outward shift = growth.
- **GDP = C + I + G + (X-M).** Counts: final goods, new housing (I), inventory changes (I), gov wages. NOT counted: used goods, intermediate, stocks/bonds, transfers, household production, illegal economy.
- **Real ≈ Nominal – Inflation.** Applies to interest rates, wages, GDP growth.
- **Inflation winners:** borrowers, gov with debt, asset owners. **Losers:** lenders, fixed-income earners, cash holders. ONLY unexpected inflation redistributes.
- **Discouraged workers:** NOT in labor force. When excluded, official rate UNDERSTATES true unemployment.
- **Bond prices and interest rates move INVERSELY.** Always.
- **Demand-pull:** AD shifts right → PL ↑, RGDP ↑, U ↓. **Cost-push:** SRAS shifts left → PL ↑, RGDP ↓, U ↑ (STAGFLATION). Only cost-push moves PL and RGDP in opposite directions.

Part 2 — Question Pattern Library

Every MCQ on the AP Macro exam fits one of these archetypes. This section names each pattern, shows the underlying technique, identifies the trap, and works through the actual practice question you got. Once you can NAME the pattern within 5 seconds of reading the question, half the work is done.

Pattern 1 — Graph Reading: PPC Feasibility

What it tests: ability to read a PPC graph and determine which combinations are inside, on, or outside the curve.

Technique: calculate slope (rise/run between two known points), then for each candidate, plug the X value into the equation and check if the Y value falls below (inside), on, or above (outside) the curve.

Trap: students panic and guess. The math is trivial — just calculate.

Worked example — your Q1

PPC passes through (0, 10) and (25, 0). Slope = $-10/25 = -0.4$. Equation: $\text{gym} = 10 - 0.4 \times \text{parks}$.

A) 5 parks, 6 gym $\rightarrow \text{max gym} = 10 - 0.4(5) = 8$. Need ≥ 6 ✓ ATTAINABLE (inside).

B) 5 parks, 8 gym $\rightarrow \text{max} = 8$. Exactly 8 ✓ ATTAINABLE (on curve).

C) 10 parks, 6 gym $\rightarrow \text{max} = 10 - 4 = 6$. Exactly 6 ✓ ATTAINABLE (on curve).

D) 15 parks, 4 gym $\rightarrow \text{max} = 10 - 6 = 4$. Exactly 4 ✓ ATTAINABLE (on curve).

E) 20 parks, 4 gym $\rightarrow \text{max} = 10 - 8 = 2$. Need $4 > 2$ ✗ UNATTAINABLE.

Answer: E.

Pattern 2 — Indeterminate Equilibrium

What it tests: the rule for what happens when BOTH demand AND supply shift simultaneously.

Technique: apply the universal both-shifts rule. SAME DIRECTION $\rightarrow$ quantity determined, price indeterminate. OPPOSITE DIRECTIONS $\rightarrow$ price determined, quantity indeterminate.

Trap: if only ONE curve shifts, both P and Q are determined. The question is asking specifically when something becomes indeterminate.

Worked example — your Q2

A) Demand $\uparrow$, Supply $\downarrow \rightarrow$ opposite directions $\rightarrow$ Price determined $\uparrow$, Quantity indeterminate. Wrong.

B) Demand $\uparrow$, Supply $\uparrow \rightarrow$ same direction (both right) $\rightarrow$ Quantity determined $\uparrow$, PRICE INDETERMINATE. ✓

C) Demand ↓, Supply ↑ → opposite directions → Price determined ↓, Quantity indeterminate. Wrong.

D) Only demand shifts → both P and Q determined. Wrong.

E) Only supply shifts → both P and Q determined. Wrong.

Answer: B.

Pattern 3 — GDP Inclusion / Exclusion

What it tests: memorization of what is and isn't counted in GDP.

Technique: apply the rule: GDP counts FINAL goods PRODUCED WITHIN the country DURING the period.

The included list (memorize)

- ✓ Final goods and services produced this period.
- ✓ NEW housing construction (counts as Investment, I).
- ✓ INVENTORY CHANGES (counts as Investment, I).
- ✓ Government wages and purchases.
- ✓ Exports.
- X Used goods (already counted when first produced).
- X Intermediate goods (avoid double-counting).
- X Stocks and bonds (financial transactions, not production).
- X Transfer payments (Social Security, welfare — no production).
- X Household production (cooking, cleaning — not market activity).
- X Illegal economy / underground economy.
- X Nonmarket activities.

Worked example — your Q3

A) Sales of stocks/bonds → financial transaction, NOT counted.

B) CHANGES IN INVENTORIES → counted as Investment. ✓

C) Changes in product quality → not directly counted (real GDP captures imperfectly).

D) Underground economy → NOT counted.

E) Nonmarket activities → NOT counted.

Answer: B.

Pattern 4 — Unemployment Rate Calculation

What it tests: formula application.

$$\text{Labor Force} = \text{Employed} + \text{Unemployed}$$

$$\text{Unemployment Rate} = (\text{Unemployed} / \text{Labor Force}) \times 100$$

CRITICAL: Population is NOT the labor force. Always use Labor Force in denominator.

Worked example — your Q4

Population = 250,000. Labor Force = 200,000. Employed = 175,000.

Unemployed = LF – Employed = 200,000 – 175,000 = 25,000.

U Rate = 25,000 / 200,000 × 100 = 12.5%.

TRAP: don't use 25,000 / 250,000 (would give 10% — answer A, wrong). The labor force is the denominator, not population.

Answer: B (12.5%).

Pattern 5 — Multiplier Calculation

What it tests: computing the change in GDP from a fiscal action, including the special case of balanced budget.

$$\text{MPC} + \text{MPS} = 1$$

$$\text{Spending Multiplier} = 1 / \text{MPS}$$

$$\text{Tax Multiplier} = - \text{MPC} / \text{MPS}$$

$$\text{Balanced Budget Multiplier} = 1 \text{ (ALWAYS)}$$

$$\Delta \text{GDP} = \Delta \text{Spending} \times \text{Spending Mult} + \Delta \text{Taxes} \times \text{Tax Mult}$$

Worked example — your Q5

MPC = 0.8, so MPS = 0.2. Spending mult = 5, Tax mult = -4.

Both G and T rise by \$100 million.

$\Delta \text{GDP} = \$100\text{M} \times 5 + \$100\text{M} \times (-4) = \$500\text{M} - \$400\text{M} = +\100 million.

OR use the shortcut: balanced budget multiplier = 1, so $\Delta \text{GDP} = \Delta G = +\100M.

Answer: C (Real GDP increases by \$100 million).

KEY INSIGHT — balanced budget mult is ALWAYS 1

Doesn't matter what MPC is. $\Delta G = \Delta T$ (same amount, same direction) $\rightarrow \Delta \text{GDP} = \Delta G$.

Why: G enters AD fully (\$100 $\rightarrow$ \$500 with mult 5). T cut increases C by $\text{MPC} \times \$100 = \80 , then $\$80 \times 5 = \400 ripple. Difference = \$100. Always.

Pattern 6 — Money Market Combined Shifts

What it tests: predicting nominal interest rate changes when BOTH MS and MD shift.

Technique: MS shift left = rate up. MS shift right = rate down. MD shift right = rate up. MD shift left = rate down. Combine using the both-shifts rule.

Worked example — your Q6

Goal: identify what raises NOMINAL interest rates.

- A) Expansionary monetary (MS right) + MD up → opposite forces → indeterminate. Wrong.
 - B) Expansionary monetary (MS right) + MD down → BOTH push rate down. Wrong.
 - C) Expansionary monetary (MS right) + MD constant → rate falls. Wrong.
 - D) Contractionary monetary (MS left) + MD up → BOTH push rate UP. ✓
 - E) Contractionary monetary (MS left) + MD down → opposite forces → indeterminate. Wrong.
- Answer: D.

Pattern 7 — Policy → AD/AS Effect

What it tests: matching policy tools to desired AD/AS movement.

To shift AD RIGHT (PL ↑, RGDP ↑, U ↓)	To shift AD LEFT (PL ↓, RGDP ↓, U ↑)
↑ G (gov spending)	↓ G
↓ T (tax cuts)	↑ T (tax increases)
↑ Transfer payments	↓ Transfer payments
Fed BUYS bonds	Fed SELLS bonds
↓ Reserve requirement	↑ Reserve requirement
↓ Discount rate	↑ Discount rate
↓ IORB (interest on reserves)	↑ IORB

Worked example — your Q7 (reduce price level in the short run)

Want: contractionary policy (AD shifts LEFT → PL falls).

- A) ↑ Transfer payments → expansionary (AD right) → PL UP. Wrong.
- B) ↑ G → expansionary (AD right) → PL UP. Wrong.
- C) ↑ IORB (interest on reserves) → contractionary monetary → MS shrinks → AD LEFT → PL DOWN. ✓
- D) ↓ Required reserve ratio → expansionary monetary → AD right → PL UP. Wrong.

E) $\downarrow$ Income tax $\rightarrow$ expansionary fiscal $\rightarrow$ AD right $\rightarrow$ PL UP. Wrong.

Answer: C.

Pattern 8 — Recessionary Gap + Policy Effect

What it tests: tracing through the standard expansionary monetary chain when economy is below potential.

Worked example — your Q8

Setup: output below potential (recessionary gap, cyclical $U > 0$). Central bank LOWERS interest rates.

Chain: lower rates $\rightarrow I \uparrow$ and durables-C $\uparrow \rightarrow$ AD shifts RIGHT $\rightarrow$ RGDP $\uparrow$.

RGDP $\uparrow \rightarrow$ cyclical unemployment DECREASES (more jobs).

RGDP $\uparrow \rightarrow$ real income INCREASES (income = real output produced).

Cyclical $U \downarrow$ + Real income $\uparrow \rightarrow$ matches choice E.

Answer: E.

Pattern 9 — Compound Policy: Two Forces, Same or Opposite Direction

What it tests: when policies REINFORCE on AD (same direction $\rightarrow$ AD definite) but OPPOSE on rates (indeterminate).

Worked example — your Q9 (both contractionary)

Contractionary fiscal: $\downarrow G$ or $\uparrow T \rightarrow$ AD LEFT. Real rate $\downarrow$ (gov stops borrowing $\rightarrow$ DLF left).

Contractionary monetary: Fed sells $\rightarrow$ MS LEFT $\rightarrow$ nominal rate $\uparrow \rightarrow$ AD LEFT. Loanable funds: SLF left $\rightarrow$ real rate $\uparrow$.

AD: BOTH push LEFT $\rightarrow$ RGDP definitely DECREASES.

Interest rates: fiscal pushes DOWN, monetary pushes UP $\rightarrow$ INDETERMINATE.

Answer: E (Real GDP will decrease, interest rates indeterminate).

KEY INSIGHT — the compound policy table

Both expansionary: AD $\uparrow\uparrow$ (definite). Real rate: forces opposite (fiscal up, monetary down) $\rightarrow$ INDETERMINATE.

Both contractionary: AD $\downarrow\downarrow$ (definite). Real rate: opposite (fiscal down, monetary up) $\rightarrow$ INDETERMINATE.

Mixed (expansionary fiscal + contractionary monetary): AD indeterminate. Real rate $\uparrow\uparrow$ (definite — both push up).

Mixed (contractionary fiscal + expansionary monetary): AD indeterminate. Real rate $\downarrow\downarrow$ (definite).

Pattern 10 — Loanable Funds Shifters

What it tests: identifying what shifts DLF (demand) vs SLF (supply).

Shifts DLF RIGHT (more borrowing)	Shifts SLF RIGHT (more saving available)
Government deficit ↑	Household savings ↑
Business confidence/optimism ↑	Foreign capital inflow
Investment tax credits	Government surplus
Consumer borrowing ↑ (homes, cars)	Higher expected inflation (real return on saving)

Worked example — your Q10

Goal: what shifts DLF right?

- A) Decrease in I → less borrowing → DLF LEFT. Wrong direction.
- B) Gov budget SURPLUS → less gov borrowing → DLF LEFT. Wrong.
- C) Firms OPTIMISTIC about future → more I → more borrowing → DLF RIGHT. ✓
- D) Domestic investors going abroad → affects supply of LF, not demand.
- E) Political instability → less I → DLF LEFT.

Answer: C.

Pattern 11 — SR vs LR Effects (the supply-side identifier)

What it tests: recognizing that ONLY supply-side improvements give RGDP↑ in SR AND PL↓ in LR.

Logic — why supply-side is the only answer

Demand-side expansions (↑G, ↓T, expansionary monetary) → AD right → RGDP ↑ AND PL ↑.
Wrong direction on PL.

Demand-side contractions → RGDP ↓. Wrong direction on RGDP.

SRAS shifts LEFT (cost-push) → RGDP ↓ AND PL ↑. Wrong on both.

SRAS shifts RIGHT (productivity, lower input costs) → RGDP ↑ AND PL ↓. ✓

If permanent → also LRAS shifts right → permanent PL ↓.

ONLY positive supply-side shocks satisfy both conditions.

Worked example — your Q11

Want: RGDP ↑ in SR AND PL ↓ in LR. Recessionary gap context.

- A) Budget surplus (contractionary fiscal) → AD left → RGDP DOWN. Wrong.
- B) Real rates rise → I ↓ → AD left → RGDP DOWN. Wrong.

C) Income tax up → AD left → RGDP DOWN. Wrong.

D) Input prices rise → SRAS LEFT → RGDP DOWN, PL UP. Wrong both.

E) PRODUCTIVITY UP → SRAS right (and LRAS right) → RGDP UP, PL DOWN. ✓

Answer: E.

Pattern 12 — Investment-to-Capital-Stock Chain

What it tests: understanding that investment incentives raise capital formation, which raises long-run output.

Worked example — your Q12 (business tax credit)

Tax credit specifically targets BUSINESSES → reduces cost of capital → INVESTMENT spending UP.

More I → more physical capital → CAPITAL STOCK UP (over time).

More capital stock → more productive capacity → REAL OUTPUT UP (LR).

Therefore: I up + capital stock up + real output up.

Answer: C.

Pattern 13 — PPC + LRAS Linked Shifts

What it tests: recognizing that PPC and LRAS represent the same concept (potential output) and shift together.

KEY INSIGHT — PPC and LRAS are linked

Both represent the economy's PRODUCTIVE CAPACITY at full employment.

Anything that raises capacity (technology, capital, labor, human capital, resources) shifts BOTH outward (PPC outward, LRAS right).

Anything that reduces capacity (resource depletion, war damage) shifts BOTH inward (PPC inward, LRAS left).

They NEVER move in opposite directions on the AP exam.

Worked example — your Q13 (technology improvement)

Tech up → more output possible from same inputs → PPC OUTWARD and LRAS RIGHT (outward).

Answer: B (Both curves shift outward).

Pattern 14 — Phillips Curve Reading

What it tests: reading inflation and unemployment values directly off the SRPC graph.

Technique: for a given inflation rate on Y-axis, trace horizontally to the SRPC, then drop down to read U on X-axis. (Or vice versa.)

Worked example — your Q14

Graph: SRPC slopes down. Visible points include (1, 8), (2, 6), (3, 4), (4, 2), (5, 1) approximately. LRPC vertical at $U = 4$.

Question: at inflation = 6%, what's U?

Trace horizontally from inflation = 6 to SRPC, then drop to X-axis. Hits at $U = 2\%$.

Answer: B (2%).

Trap — don't confuse SRPC and LRPC

SRPC tells you the trade-off in the SHORT run.

LRPC is vertical at NRU — represents the long run with no trade-off.

Question asks about CURRENT unemployment → use SRPC.

If question asks about long-run U after expectations adjust → use LRPC (vertical line).

Pattern 15 — Forex / Budget Deficit Chain

What it tests: tracing fiscal policy through loanable funds, real interest rate, capital flows, to currency value.

KEY CHAIN — deficit to currency (memorize)

Step 1: Government deficit $\uparrow$ → Gov borrows more → DLF shifts RIGHT.

Step 2: Real interest rate UP.

Step 3: Higher US real rates attract foreign capital → CAPITAL INFLOW.

Step 4: Foreigners need DOLLARS to buy US assets → D_{USD} shifts RIGHT.

Step 5: Dollar APPRECIATES.

Result: deficit $\uparrow$ → rates $\uparrow$ → dollar appreciates.

Worked example — your Q15

US budget deficit increases → real rates rise → capital inflow → dollar appreciates.

A) Appreciate because rates increase. ✓

B) Depreciate + rates decrease. Wrong direction on rates.

C) Appreciate + rates decrease. Inconsistent (the appreciation comes FROM higher rates).

D) Depreciate + rates increase. Inconsistent.

E) No change. Wrong.

Answer: A.

Part 3 — Calculation Fluency Block

Every calculation the AP exam tests, drilled with worked examples and practice problems. Target: under 30 seconds per calculation. Practice problems are at the end of each section with answers.

3.1 — GDP, Real GDP, Deflator, CPI

$$\begin{aligned}\text{GDP} &= C + I + G + (X - M) \\ \text{Real GDP} &= (\text{Nominal GDP} / \text{GDP Deflator}) \times 100 \\ \text{GDP Deflator} &= (\text{Nominal GDP} / \text{Real GDP}) \times 100 \\ \text{CPI} &= (\text{Cost of basket now} / \text{Cost in base year}) \times 100 \\ \text{Inflation Rate} &= ((\text{CPI}_2 - \text{CPI}_1) / \text{CPI}_1) \times 100\end{aligned}$$

Worked example A — Real GDP from Nominal + Deflator

Nominal GDP 2024 = \$1,500B. GDP Deflator 2024 = 125.

Real GDP = $(1,500 / 125) \times 100 = 12 \times 100 = \$1,200\text{B}$.

Interpretation: in base-year prices, the economy produced \$1,200B worth of stuff.

Worked example B — Inflation between two years

CPI 2023 = 240. CPI 2024 = 252.

Inflation 2023→2024 = $(252 - 240) / 240 \times 100 = 12 / 240 \times 100 = 5\%$.

Sanity check: 5% inflation means prices rose 5% on average.

Worked example C — Real GDP from Fehran (your FRQ #1)

Nominal GDP 2012 = \$225,000. GDP Deflator 2012 = 125.

Real GDP 2012 = $(225,000 / 125) \times 100 = 1,800 \times 100 = \$180,000$.

Inflation 2011→2012: deflator went from 100 to 125. Inflation = $(125 - 100) / 100 \times 100 = 25\%$.

Practice problems

1. Nominal GDP = \$2,400B. Deflator = 120. Find Real GDP.
2. Real GDP = \$5,000B. Nominal GDP = \$6,000B. Find the GDP deflator.
3. CPI rose from 180 to 198. Find inflation rate.
4. Country has Nominal GDP \$1,000B and Real GDP \$800B. Find the deflator.
5. CPI 2020 = 200, CPI 2021 = 210, CPI 2022 = 220.5. Find inflation 2020→2021 and 2021→2022.

Answers

- 1) Real GDP = $(2,400 / 120) \times 100 = \$2,000\text{B}$.
- 2) Deflator = $(6,000 / 5,000) \times 100 = 120$.

3) Inflation = $(198-180)/180 \times 100 = 18/180 \times 100 = 10\%$.

4) Deflator = $(1,000/800) \times 100 = 125$.

5) 2020→2021: $(210-200)/200 = 5\%$. 2021→2022: $(220.5-210)/210 \times 100 = 5\%$.

3.2 — Per Capita Real GDP / Standard of Living

$$\text{Real GDP per capita} = \text{Real GDP} / \text{Population}$$

Why this matters: Standard of living depends on REAL GDP PER CAPITA, not just real GDP. If real GDP grows 5% but population grows 5%, standard of living is unchanged.

Worked example — Fehran standard of living (your FRQ #1, Part C)

2011: Nominal GDP \$150,000. Deflator 100. Real GDP = $(150,000/100) \times 100 = \$150,000$. Population 100.

Real GDP per capita 2011 = $150,000/100 = \$1,500$.

2012: Real GDP = \$180,000 (calculated earlier). Population 120.

Real GDP per capita 2012 = $180,000/120 = \$1,500$.

Standard of living STAYED THE SAME — real GDP grew (150K→180K, +20%), but population grew at the same rate (100→120, +20%).

Practice

1. Country had real GDP \$400B, population 40M. Five years later: real GDP \$500B, population 50M. What happened to standard of living?
2. Real GDP rises 8%. Population rises 3%. Approximately what happens to per-capita real GDP?

Answers

- 1) Per capita: $\$10,000 \rightarrow \$10,000$. Unchanged.
- 2) Per capita $\approx 8\% - 3\% = \sim 5\%$ increase.

3.3 — Unemployment Rate, LFPR

Labor Force = Employed + Unemployed (actively seeking)

Unemployment Rate = $(\text{Unemployed} / \text{Labor Force}) \times 100$

LFPR = $(\text{Labor Force} / \text{Working-Age Population}) \times 100$

Natural Rate = Frictional + Structural

Trap — population vs labor force

Population includes children, retirees, those not seeking work.

Working-age population = adults 16+, NOT in institutions.

Labor Force = subset of working-age population WHO ARE WORKING OR ACTIVELY SEEKING.

Discouraged workers are NOT in the labor force.

U rate denominator is ALWAYS labor force, never population.

Worked example — your Q4

Population 250,000. Labor Force 200,000. Employed 175,000.

Unemployed = $200,000 - 175,000 = 25,000$.

U Rate = $25,000 / 200,000 \times 100 = 12.5\%$.

(Don't divide by 250,000 — that gives 10%, the trap answer.)

Worked example — discouraged worker effect

Initial: LF = 150M, Employed = 140M, Unemployed = 10M. U Rate = $10/150 = 6.67\%$.

5M discouraged workers re-enter the labor force, still seeking.

New LF = 155M, Unemployed = 15M. U Rate = $15/155 = 9.68\%$.

Official rate ROSE even though no real worsening — these people were always jobless. Just now counted.

Practice

- Population 500,000. Working-age population 380,000. Labor force 280,000. Employed 252,000. Find U rate and LFPR.
- LF = 100M. NRU = 5%. Cyclical U = 3%. Find total unemployment and the unemployment rate.

Answers

1) Unemployed = $280K - 252K = 28K$. U rate = $28/280 \times 100 = 10\%$. LFPR = $280/380 \times 100 \approx 73.7\%$.

2) Total U = NRU + Cyclical = $5\% + 3\% = 8\%$. Unemployed people = 8M (8% of 100M).

3.4 — Multipliers

$$MPC + MPS = 1$$

$$\text{Spending Multiplier} = 1/MPS = 1/(1-MPC)$$

$$\text{Tax Multiplier} = -MPC/MPS$$

$$\text{Balanced Budget Multiplier} = 1 \text{ (always)}$$

$$\Delta GDP = \Delta \text{Spending} \times \text{Spending Mult}$$

$$\Delta GDP = \Delta \text{Taxes} \times \text{Tax Mult}$$

$$\Delta \text{GDP} = \Delta \text{Spending} \times \text{Spending Mult} + \Delta \text{Taxes} \times \text{Tax Mult (combined)}$$

The five values to memorize cold

MPC = 0.5 → Spending mult 2, Tax mult -1

MPC = 0.6 → Spending mult 2.5, Tax mult -1.5

MPC = 0.75 → Spending mult 4, Tax mult -3

MPC = 0.8 → Spending mult 5, Tax mult -4

MPC = 0.9 → Spending mult 10, Tax mult -9

Pattern: Tax mult = Spending mult - 1 (in absolute value).

MULTIPLY vs DIVIDE — the question type matters

Type A: "What's the change in GDP from \$200B in spending?" → MULTIPLY. $\Delta \text{GDP} = \$200\text{B} \times \text{mult}$.

Type B: "What change in spending closes a \$200B gap?" → DIVIDE. $\Delta \text{Spending} = \$200\text{B} / \text{mult}$.

Read the question slowly. Most lost points come from this.

Worked example A — calibrate to close gap

Recessionary gap = \$400B. MPC = 0.75 → spending mult = 4, tax mult = -3.

Option 1 (G alone): $\Delta G \times 4 = \$400\text{B} \rightarrow \Delta G = +\100B .

Option 2 (T cut alone): $\Delta T \times (-3) = \$400\text{B} \rightarrow \Delta T = -\133.3B (cut taxes by \$133B).

Tax cut needs to be LARGER than spending increase (because part is saved).

Worked example B — balanced budget (your Q5)

MPC = 0.8 → spending mult = 5, tax mult = -4.

$\Delta G = +\$100\text{M}$, $\Delta T = +\$100\text{M}$.

$\Delta \text{GDP} = 100 \times 5 + 100 \times (-4) = 500 - 400 = +\100M .

Shortcut: balanced budget mult = 1, so $\Delta \text{GDP} = \Delta G = \100M .

Practice

1. MPC = 0.75. Government raises spending by \$300B. Find ΔGDP .
2. MPC = 0.8. Output gap is \$1,000B (recessionary). What change in taxes closes it?
3. MPC = 0.6. Government raises G by \$400B and T by \$400B. Find ΔGDP .
4. Inflationary gap of \$200B. MPC = 0.75. What change in G closes it?

Answers

1) $\Delta \text{GDP} = 300 \times 4 = +\$1,200\text{B}$.

2) Tax mult = -4. $\Delta T \times (-4) = \$1,000\text{B} \rightarrow \Delta T = -\250B (cut by \$250B).

3) Balanced budget → $\Delta \text{GDP} = +\$400\text{B}$.

4) Spending mult = 4. $\Delta G \times 4 = -\$200B \rightarrow \Delta G = -\$50B$ (cut spending by \$50B).

3.5 — Money Multiplier and Banking

$$\text{Required Reserves} = \text{Deposits} \times \text{RRR}$$

$$\text{Excess Reserves} = \text{Total Reserves} - \text{Required Reserves}$$

$$\text{Money Multiplier} = 1 / \text{RRR}$$

$$\text{Maximum } \Delta \text{Money Supply (from new lending)} = \text{Excess Reserves} \times \text{Money Mult}$$

Worked example — full chain

RRR = 10%. Bank receives a NEW \$1,000 deposit.

Required reserves = $1,000 \times 0.10 = \$100$.

Excess reserves created = $\$1,000 - \$100 = \$900$.

Money multiplier = $1/0.10 = 10$.

Maximum new loans = $\$900 \times 10 = \$9,000$.

Max total Δ MS (including original deposit, if it's new to system) = $\$1,000 \times 10 = \$10,000$.

READ THE QUESTION CAREFULLY

"Maximum increase in money supply from this deposit" $\rightarrow$ use full deposit $\times$ multiplier.

"Maximum new loans" or "new money created" $\rightarrow$ excess reserves $\times$ multiplier.

"Maximum increase if bank loans out all excess" $\rightarrow$ excess $\times$ multiplier.

Practice

1. RRR = 20%. Bank has \$10,000 deposits, \$4,000 total reserves. Find required, excess, money mult, max new loans.
2. RRR = 25%. New deposit of \$2,000. Find max new lending throughout the system.

Answers

1) Required = $10K \times 0.20 = \$2K$. Excess = $4K - 2K = \$2K$. Mult = $1/0.20 = 5$. Max new loans = $2K \times 5 = \$10K$.

2) Required = $2K \times 0.25 = \$500$. Excess = $\$1,500$. Mult = 4. Max new loans = $\$1,500 \times 4 = \$6,000$.

3.6 — Real Interest Rate (Fisher)

$$\text{Real interest rate} \approx \text{Nominal interest rate} - \text{Inflation rate}$$

(Fisher equation, approximation)

Worked example

Nominal rate = 7%. Inflation = 4%. Real rate $\approx$ 3%.

Borrower scenario: borrows at 7% nominal. Inflation reduces the real burden.

→ Real cost of borrowing is only 3%.

Lender scenario: receives 7% nominal but loses 4% to inflation. Real return only 3%.

Trap — confusing real and nominal in interest rate questions

INVESTMENT decisions depend on REAL rate (the actual cost of borrowing in purchasing-power terms).

If nominal rises 2% but inflation rises 4%, real rate FELL 2% → investment INCREASES.

Always identify which rate the question is about.

Practice

1. Nominal = 8%, inflation = 3%. Find real rate.
2. Real = 4%, inflation = 6%. Find nominal.
3. Nominal rises from 5% to 7%. Inflation rises from 2% to 5%. What happens to real rate?

Answers

- 1) Real $\approx$ 5%.
- 2) Nominal $\approx$ 10%.
- 3) Real was 3%, now 2%. Real rate FELL 1 percentage point.

3.7 — Quantity Theory of Money

$$MV = PQ$$

M = Money supply

V = Velocity (rate at which money circulates)

P = Price level

Q = Real output

Approximation: $\% \Delta M \approx \% \Delta P + \% \Delta Q$ (when V constant)

Solving for inflation: $\% \Delta P \approx \% \Delta M - \% \Delta Q$

Worked example

Money supply grows 8%/year. Real output grows 2%/year. Velocity constant.

Inflation $\approx$ 8% - 2% = 6%.

Country with 30% money growth + 2% real growth → ~28% inflation. Hyperinflation territory.

Practice

1. M grows 5%, V constant, Q grows 3%. Find inflation.
2. Inflation is 10%, real growth is 2%, V constant. Find money growth.

Answers

- 1) Inflation $\approx 5\% - 3\% = 2\%$.
- 2) M growth $\approx 10\% + 2\% = 12\%$.

3.8 — Budget Deficit and National Debt

Budget Deficit (year) = Government Spending - Tax Revenue
(negative = surplus)

National Debt = Cumulative sum of past deficits (minus surpluses)

When deficit > 0 : government must BORROW $\rightarrow$ debt RISES
When surplus > 0 : government can pay down debt $\rightarrow$ debt FALLS

Worked example — Fehran (your FRQ #1, Part D)

2011: $G = \$25K$, $T = \$25K \rightarrow$ Balanced (deficit = 0).

2012: $G = \$35K$, $T = \$30K \rightarrow$ Deficit = $\$5K$.

From 2011 to 2012: Fehran went from balanced to running a deficit.

National debt INCREASES by $\$5K$ (the amount of the new deficit).

Plus any prior debt continues to exist; deficits ADD to it.

Practice

1. Year 1: $G = \$500B$, $T = \$480B$. Year 2: $G = \$520B$, $T = \$510B$. Compare deficits and debt change.

Answer

Year 1 deficit = $\$20B$. Year 2 deficit = $\$10B$ (smaller).

But debt still rose: total addition = $\$20B + \$10B = \$30B$ over 2 years.

Part 4 — Indeterminacy Mastery

The 'both curves shift' rule is among the most-tested concepts on the AP Macro exam — Q2 and Q9 in your practice both hinge on it. Master this single rule and you'll solve a class of questions that trips up most students.

4.1 — The Universal Rule

THE RULE — applies to every market

When BOTH curves shift in the SAME DIRECTION:

- Quantity is DETERMINED (you know which way Q moves).
- Price is INDETERMINATE (depends on relative magnitudes).

When BOTH curves shift in OPPOSITE DIRECTIONS:

- Price is DETERMINED.
- Quantity is INDETERMINATE.

Same → Q | Opposite → P. Memorize.

4.2 — Rule Applied: Goods Market

Demand	Supply	P	Q
↑ (right)	↑ (right)	Indeterminate	Definitely ↑
↓ (left)	↓ (left)	Indeterminate	Definitely ↓
↑ (right)	↓ (left)	Definitely ↑	Indeterminate
↓ (left)	↑ (right)	Definitely ↓	Indeterminate

4.3 — Rule Applied: Money Market (Q6 territory)

MS	MD	Nominal Rate	Q of Money
↑ (right) — expansionary	↑ (right)	Indeterminate	Definitely ↑
↓ (left) — contractionary	↓ (left)	Indeterminate	Definitely ↓
↑ (right) — expansionary	↓ (left)	Definitely ↓	Indeterminate
↓ (left) — contractionary	↑ (right)	Definitely ↑	Indeterminate

Reading the rule for money market

MS goes RIGHT when Fed expands (buys bonds, lowers IORB, etc.). Pushes nominal rate DOWN.

MD goes RIGHT when income or PL rises. Pushes nominal rate UP.

If both expansionary AND income up: rate goes 'down then up' → indeterminate.

If contractionary AND income up: BOTH push rate up → definitely up. (This is your Q6 answer D.)

4.4 — Rule Applied: Compound Fiscal + Monetary on Real Rate (Q9 territory)

Compound policy on real interest rate

Expansionary FISCAL: gov borrows → DLF right → real rate UP.

Expansionary MONETARY: Fed adds liquidity → SLF right → real rate DOWN.

Contractionary FISCAL: gov stops borrowing → DLF left → real rate DOWN.

Contractionary MONETARY: Fed drains liquidity → SLF left → real rate UP.

Fiscal	Monetary	Effect on Real Rate	Effect on RGDP
Expansionary	Expansionary	INDETERMINATE (forces opposite)	Definitely ↑↑
Contractionary	Contractionary	INDETERMINATE (forces opposite)	Definitely ↓↓
Expansionary	Contractionary	Definitely ↑↑ (both up)	INDETERMINATE
Contractionary	Expansionary	Definitely ↓↓ (both down)	INDETERMINATE

Worked example — your Q9 answered systematically

Both contractionary fiscal AND monetary.

Real GDP: both push AD LEFT → DEFINITELY DOWN.

Real rate: fiscal pushes DOWN (gov stops borrowing), monetary pushes UP (Fed drains liquidity) → INDETERMINATE.

Match to choices: Real GDP definitely down + interest rates indeterminate.

Answer: E ('Real GDP will decrease, change in interest rates indeterminate').

4.5 — Rule Applied: Loanable Funds

DLF (borrowing demand)	SLF (saving supply)	Real Rate	Q of LF
↑ (right)	↑ (right)	Indeterminate	Definitely ↑
↓ (left)	↓ (left)	Indeterminate	Definitely ↓
↑ (right)	↓ (left)	Definitely ↑	Indeterminate
↓ (left)	↑ (right)	Definitely ↓	Indeterminate

4.6 — Rule Applied: Forex

D for currency	S for currency	Exchange Rate	Q of currency traded
↑	↑	Indeterminate	Definitely ↑
↓	↓	Indeterminate	Definitely ↓
↑	↓	Definitely ↑ (appreciate)	Indeterminate
↓	↑	Definitely ↓ (depreciate)	Indeterminate

4.7 — Recognition Drill

For each scenario, identify which variable is determined and which is indeterminate:

1. Income rises (MD up) AND Fed buys bonds (MS up). Effect on nominal rate?
2. Government runs deficit (DLF up) AND foreign capital inflows (SLF up). Effect on real rate?
3. Bad weather destroys crops (S left) AND consumer income rises for normal good (D up). Effect on price and quantity?
4. Expansionary monetary (real rate down) AND positive supply shock (SRAS right). Effect on PL? Effect on RGDP?
5. Contractionary fiscal AND expansionary monetary. Effect on real rate? Effect on AD?

Answers

- 1) MS up (rate down) + MD up (rate up) = INDETERMINATE rate. Q of money definitely up.
- 2) DLF up (rate up) + SLF up (rate down) = INDETERMINATE real rate. Q of LF definitely up.
- 3) Opposite directions (S left, D right): Price DEFINITELY UP. Quantity INDETERMINATE.
- 4) AD right (rate down → I up) + SRAS right: PL — AD pushes up, SRAS pushes down → INDETERMINATE PL. RGDP definitely UP (both push right).

5) Real rate: fiscal down + monetary down → DEFINITELY DOWN. AD: fiscal left + monetary right → INDETERMINATE.

Part 5 — Unit 3 Deep: AD/AS as the Hub

Exam weight: 17–27%. The most-graphed model on the exam. Every other model feeds into or out of it.

5.1 — Why AD slopes down (the three effects)

These explain the SHAPE of AD. They cause MOVEMENT ALONG AD when PL changes — they are NOT shifters.

1. **Wealth effect:** higher PL erodes the real value of nominal assets (cash, bonds). People feel poorer → spend less.
2. **Interest rate effect:** higher PL → people need more money for transactions → MD ↑ → nominal rate ↑ → I and durables-C ↓.
3. **Exchange rate effect:** higher US PL → US goods more expensive than foreign → X ↓, M ↑ → (X-M) ↓.

Critical distinction

These three effects = WHY AD slopes down. They explain the curve's shape.

They are NOT shifters. They cause movement along AD when PL changes due to AS shifting.

AD shifts only when C, I, G, or (X-M) changes for reasons other than a PL change.

5.2 — AD shifters (CIGXM framework)

Component	Increases (AD right)	Decreases (AD left)
C — Consumption	↑ Consumer confidence/wealth, ↓ taxes, ↑ transfers, ↓ rates, ↓ household debt	Reverse
I — Investment	↑ Business confidence, ↓ real rate, ↓ business taxes, ↑ technology, investment tax credit	Reverse
G — Gov spending	↑ G (expansionary fiscal)	↓ G (contractionary)
X – M — Net exports	↑ Foreign income, dollar DEPRECIATES (US goods cheaper abroad)	Foreign income ↓, dollar APPRECIATES

5.3 — SRAS shifters

SRAS slopes UP because in the short run, wages and many input prices are STICKY. When PL rises, firms get more revenue but fixed input costs → margins rise → expand output.

- **Input prices** (wages, oil, raw materials) ↑ → SRAS LEFT.

- **Productivity / technology** $\uparrow \rightarrow$ SRAS RIGHT.
- **Expected inflation** $\uparrow \rightarrow$ workers demand higher wages $\rightarrow$ SRAS LEFT.
- **Business taxes / regulation** $\uparrow \rightarrow$ SRAS LEFT.
- **Subsidies** $\uparrow \rightarrow$ SRAS RIGHT.
- **Negative supply shock** (disaster, war, oil spike) $\rightarrow$ SRAS LEFT.

5.4 — LRAS — vertical at potential GDP

LRAS is VERTICAL at the economy's potential output. Long-run output is **INDEPENDENT** of PL because in the long run all wages and prices are flexible.

LRAS = potential GDP = full-employment GDP

All four phrases mean the same thing: the level of RGDP at which $U = \text{NRU}$.

LRAS shift = long-run economic growth. PPC shift outward = long-run growth.

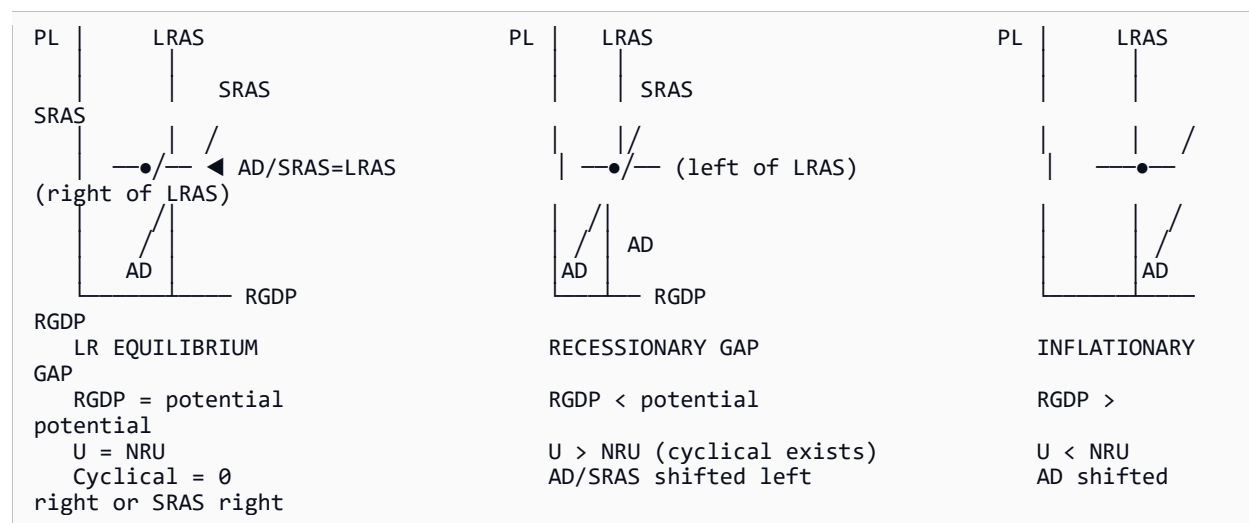
On the LRAS, $U = \text{NRU}$. Cyclical unemployment = 0.

PPC and LRAS shift **TOGETHER** (your Q13 tested this directly).

LRAS shifters

- **Quantity of resources:** more labor (immigration, LFPR), more capital (investment), more natural resources.
- **Quality of resources:** better education (human capital), better technology.
- **Productivity gains** from infrastructure, R&D, supply-side reforms.

5.5 — The three equilibria



5.6 — Self-correction (classical view)

Recessionary gap → self-correction

1. High unemployment puts downward pressure on wages.
2. Lower wages = lower input costs.
3. SRAS shifts RIGHT.
4. New equilibrium back at LRAS. PL is LOWER, RGDP at potential.

Inflationary gap → self-correction (your FRQ #2 Part B)

1. Tight labor market puts upward pressure on wages.
2. Higher wages = higher input costs.
3. SRAS shifts LEFT.
4. New equilibrium back at LRAS. PL is HIGHER, RGDP at potential.

This is exactly what's happening in Zeetoland: actual U (5%) below NRU (6%) → inflationary gap → wages rise → SRAS left → return to potential at higher PL.

Why Keynesians disagree

Keynesian view: wages are STICKY DOWNWARD. Workers won't accept wage cuts; firms can't easily fire and rehire at lower wages.

→ Recessionary gaps can persist for years without active policy.

→ Justifies expansionary fiscal/monetary policy.

AP exam may ask you to compare classical (self-correcting) vs Keynesian (intervention needed).

Know both views.

5.7 — Multipliers and Fiscal Policy

$$\text{MPC} + \text{MPS} = 1$$

$$\text{Spending Multiplier} = 1/\text{MPS}$$

$$\text{Tax Multiplier} = -\text{MPC}/\text{MPS}$$

$$\text{Balanced Budget Multiplier} = 1$$

$$\Delta \text{GDP} = \Delta \text{Spending} \times \text{Spending Mult} \quad \text{OR} \quad \Delta \text{Taxes} \times \text{Tax Mult}$$

Worked example — calibrating policy

Recessionary gap = \$400B. MPC = 0.75 (spending mult = 4, tax mult = -3).

Option A — Increase G: $\Delta G \times 4 = \$400\text{B} \rightarrow \Delta G = \100B .

Option B — Cut T: $\Delta T \times (-3) = \$400\text{B} \rightarrow \Delta T = -\133.3B .

Tax cuts must be LARGER than spending increases — only MPC of the cut enters the spending stream initially.

5.8 — Fiscal Policy → Loanable Funds → Crowding Out

Goal	Tool	Effect on AD	Effect on real interest rate
Stimulate economy	$\uparrow G$ or $\downarrow T$	AD shifts RIGHT	$\uparrow$ (gov borrowing → DLF right) — CROWDING OUT
Cool economy	$\downarrow G$ or $\uparrow T$	AD shifts LEFT	$\downarrow$ (less gov borrowing)

5.9 — Automatic Stabilizers

Definition: policies that automatically counteract the business cycle WITHOUT new legislation.

- **Progressive income tax:** boom → more tax revenue, dampens boom. Recession → less tax taken.
- **Unemployment insurance:** recession → more transfers, dampens contraction.
- **Welfare / SNAP:** similar countercyclical effect.

Stabilizers shrink the business cycle

Recessions are less severe; booms are less hot. Stabilizers reduce the EFFECTIVE multiplier.

Automatic ≠ discretionary. Discretionary = new laws. Automatic = built into existing programs.

Part 6 — Unit 4 Deep: Money, Banking, and Two Interest Rates

Exam weight: 18–23%. Stated weak point. Spend the time. The single most-confused area: money market vs loanable funds market.

6.1 — The Two Interest Rates — never confuse them

Money Market vs Loanable Funds — the line that solves Unit 4

MONEY MARKET:

- Y-axis: NOMINAL interest rate.
- X-axis: Quantity of money.
- Curves: MD (downward), MS (vertical).
- Driven by: Federal Reserve. MS shifts when Fed acts. MD shifts when GDP or PL changes.
- Question keywords: 'Fed,' 'open-market operations,' 'M1, M2,' 'discount rate,' 'reserve requirement,' 'liquidity.'

LOANABLE FUNDS MARKET:

- Y-axis: REAL interest rate.
- X-axis: Quantity of loanable funds.
- Curves: DLF (downward), SLF (upward).
- Driven by: Borrowers and savers. SLF shifts when saving changes. DLF shifts when government borrows or business confidence changes.
- Question keywords: 'savings,' 'investment,' 'government deficit,' 'crowding out,' 'capital inflow.'

Identify which market a question is about within 5 seconds. Most of Unit 4 becomes solvable from there.

6.2 — Bond Prices and Interest Rates (the inverse relationship)

Why bond prices and rates move INVERSELY

A bond promises FIXED future payments. Suppose a \$1,000 bond pays \$50/year forever.

If bond costs \$1,000 today, $\text{yield} = 50/1,000 = 5\%$.

If new bonds are issued at 10%, no one pays \$1,000 for the old 5% bond. Its PRICE FALLS until effective yield matches 10%. $\sim \$500 \rightarrow 50/500 = 10\%$.

Bond prices $\uparrow \Leftrightarrow$ interest rates $\downarrow$.

Bond prices $\downarrow \Leftrightarrow$ interest rates $\uparrow$.

FED BUYS BONDS: bond demand rises $\rightarrow$ bond prices $\uparrow \rightarrow$ interest rates $\downarrow$. Expansionary.

FED SELLS BONDS: bond supply rises $\rightarrow$ bond prices $\downarrow \rightarrow$ interest rates $\uparrow$. Contractionary.

CONTRACTIONARY MONETARY ACTION (your FRQ #2 Part D-i): bond prices FALL.

6.3 — Money: Functions and Categories

Three functions of money

1. **Medium of exchange** (accepted in trade — solves the double coincidence of wants).
2. **Store of value** (preserves purchasing power across time — fails under high inflation).
3. **Unit of account** (common measure of value — we price everything in dollars).

Money supply categories

Category	Includes	Liquidity
M0 (monetary base)	Physical currency in circulation + bank reserves at Fed	Most liquid
M1	M0 + checking deposits + traveler's checks	Highly liquid
M2	M1 + savings, small CDs, money market mutual funds	Less liquid

What's NOT money

Stocks and bonds — financial assets, not in M1 or M2.

Credit cards — a form of borrowing.

Gold — no longer money; it's a commodity asset.

6.4 — Banking and Money Multiplier

Banks operate on a balance sheet: Assets = Liabilities + Net Worth.

ASSETS	LIABILITIES + EQUITY
Required reserves (% of deposits)	Demand deposits (checking accounts)
Excess reserves (above required)	Savings deposits
Loans outstanding	Borrowed funds (from Fed, other banks)
Securities (gov bonds, etc.)	Owner's equity (net worth)

$$\text{Required reserves} = \text{Deposits} \times \text{RRR}$$

$$\text{Excess reserves} = \text{Total reserves} - \text{Required reserves}$$

$$\text{Money Multiplier} = 1 / \text{RRR}$$

$$\text{Maximum } \Delta \text{Money Supply} = \text{Initial excess reserves} \times \text{Money Multiplier}$$

Worked example — money multiplier in action

RRR = 10%. New \$1,000 deposit.

Required = $\$1,000 \times 0.10 = \100 .

Excess = $\$1,000 - \$100 = \$900$.

Bank lends \$900. Borrower deposits it elsewhere.

Next bank: required \$90, lends \$810. And so on.

Maximum new money created from lending = $\$900 \times (1/0.10) = \$9,000$.

Total Δ MS (including original deposit if new to system) = $\$1,000 \times 10 = \$10,000$.

READ: 'maximum increase in MS' $\rightarrow$ full deposit $\times$ mult. 'Maximum new loans' $\rightarrow$ excess $\times$ mult.

6.5 — The Money Market

MS is VERTICAL because the Fed sets it directly (perfectly inelastic).

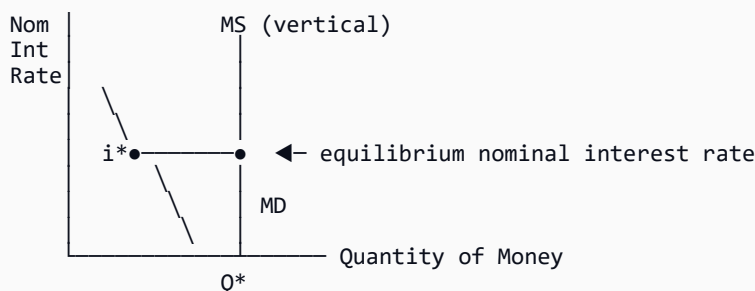
MD slopes downward because the nominal interest rate is the OPPORTUNITY COST of holding money.

MD shifters

- **Real GDP / income $\uparrow$** $\rightarrow$ more transactions $\rightarrow$ MD RIGHT.
- **Price level $\uparrow$** $\rightarrow$ each transaction needs more money $\rightarrow$ MD RIGHT.

MS shifters (the Fed's tools)

- **Fed BUYS bonds** $\rightarrow$ MS RIGHT. Most-used tool.
- **Fed SELLS bonds** $\rightarrow$ MS LEFT.
- **Reserve requirement $\downarrow$** $\rightarrow$ MS RIGHT (more lending capacity).
- **Discount rate $\downarrow$** $\rightarrow$ MS RIGHT (banks borrow more from Fed).
- **IORB $\downarrow$** $\rightarrow$ MS effectively RIGHT (banks lend rather than hold reserves).



When MS shifts right (Fed buys bonds): i^* falls.
 When MD shifts right (income or PL rises): i^* rises.

6.6 — The Loanable Funds Market

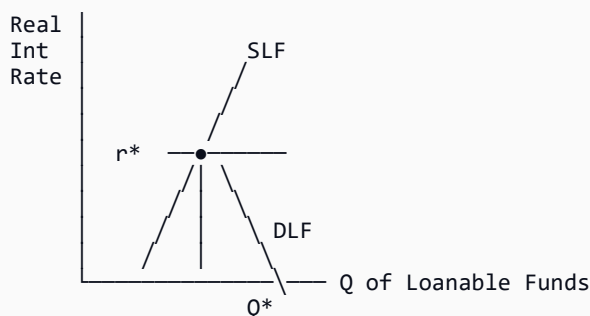
DLF (downward) comes from BORROWERS — businesses, gov, households. SLF (upward) comes from SAVERS.

DLF shifters

- **Government deficit / borrowing** $\uparrow \rightarrow$ DLF RIGHT (CROWDING OUT).
- **Business confidence/optimism** $\uparrow \rightarrow$ DLF RIGHT. (Your Q10 tested this.)
- **Investment tax credits** $\rightarrow$ DLF RIGHT. (Your Q12 — incentive raises borrowing demand.)
- **Consumer big-ticket borrowing** $\uparrow$ (homes, cars) $\rightarrow$ DLF RIGHT.

SLF shifters

- **Household savings rate** $\uparrow \rightarrow$ SLF RIGHT.
- **Foreign capital inflow** $\rightarrow$ SLF RIGHT.
- **Government surplus** $\rightarrow$ SLF RIGHT (gov saves rather than borrows).
- **Higher expected future income** $\rightarrow$ SLF LEFT (less saving needed now).

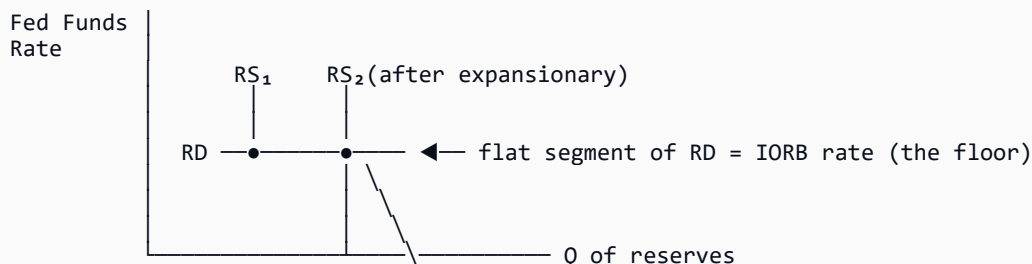


Gov deficit $\rightarrow$ DLF right $\rightarrow r^*$ up $\rightarrow$ private I down (crowding out).
 Foreign capital inflow $\rightarrow$ SLF right $\rightarrow r^*$ down.

6.7 — Reserve Market (Modern Fed)

Since 2008, the Fed operates with ABUNDANT reserves. The reserve market graph has changed:

- **Reserve Demand (RD)** is STEPPED — flat at the IORB rate (the floor), then steeply downward when reserves are scarce.
- **Reserve Supply (RS)** is vertical (Fed sets it).
- **Equilibrium** normally happens on the FLAT segment (ample reserves).



EXPANSIONARY: Fed lowers IORB OR shifts RS right → policy rate falls.
 CONTRACTIONARY: Fed raises IORB OR shifts RS left → policy rate rises.

6.8 — The Transmission Mechanism (THE most important chain)

EXPANSIONARY MONETARY POLICY — chain (memorize)

1. Fed BUYS bonds (or lowers IORB).
 2. MS shifts RIGHT in money market → nominal interest rate ↓.
 3. SLF shifts RIGHT in loanable funds → real interest rate ↓.
 4. Lower real rate → $I \uparrow$ and durables- $C \uparrow$.
 5. AD shifts RIGHT in AD/AS → $PL \uparrow$, $RGDP \uparrow$, $U \downarrow$.
 6. (Open economy) Lower US rates → capital outflow → S_USD right → dollar DEPRECIATES → $X \uparrow$, $M \downarrow$ → AD ↑ MORE.
 7. On Phillips curve: MOVE up-left along SRPC (inflation up, U down).
- If you can recite this chain in your sleep, half of Unit 4 FRQ questions are automatic.

CONTRACTIONARY MONETARY POLICY — mirror chain (your FRQ #2 territory)

1. Fed SELLS bonds (or raises IORB).
2. MS shifts LEFT → nominal $i \uparrow$. Bond prices FALL (Part D-i answer).
3. Real rate ↑.
4. $I \downarrow$, durables- $C \downarrow$ → AD shifts LEFT → $PL \downarrow$, $RGDP \downarrow$, $U \uparrow$. (Real output falls — Part D-ii.)
5. (Open economy) Higher US rates → capital INFLOW (Part D-iii: net financial capital flows TO Zeetoland) → D_USD right → dollar APPRECIATES (Part E).
6. Stronger currency → exports DECREASE (Part F).
7. Phillips: MOVE down-right along SRPC.

6.9 — Common Unit 4 Traps

Trap 1 — Money market vs Loanable funds

Money market = nominal rate, Fed-driven.

Loanable funds = real rate, saver/borrower-driven.

Crowding out happens in LOANABLE FUNDS, not money market.

Trap 2 — 'What is INVESTMENT in macro?'

Macro 'investment' = physical capital + new homes + inventory changes.

NOT buying stocks or bonds (financial transaction).

When the question asks about investment, they mean BUSINESS investment in plant/equipment + housing.

Trap 3 — Initial deposit vs initial excess reserves

\$1,000 deposit, RRR = 10%:

- Excess reserves CREATED = \$900.
- Maximum NEW LOANS = $\$900 \times 10 = \$9,000$.
- Maximum total ΔMS (including the deposit) = $\$1,000 \times 10 = \$10,000$.

Read the question — what exactly are they asking for?

Trap 4 — Real vs nominal rate effects on investment

Investment is driven by the REAL interest rate.

If nominal rate rises 2% but inflation rises 4%, REAL rate FELL 2% → investment INCREASES.

Always identify which rate the question is about.

Part 7 — Unit 5 Deep: Long-Run Consequences (highest weight)

Exam weight: 20–30%. HIGHEST-WEIGHTED unit AND your stated weak point. This unit is where Units 3 and 4 stop being separate — Unit 5 is the synthesis.

7.1 — Combined Fiscal and Monetary Policy

Fiscal and monetary policy can either reinforce each other or work against each other. Both affect AD. Both affect interest rates — but in OPPOSITE ways. (This is exactly Q9 territory.)

Fiscal	Monetary	Net AD effect	Net real interest rate effect
Expansionary	Expansionary	Strong ↑ (compounded)	INDETERMINATE (forces offset)
Expansionary	Contractionary	INDETERMINATE (offset)	Strong ↑ (both up)
Contractionary	Contractionary	Strong ↓	INDETERMINATE (forces offset)
Contractionary	Expansionary	INDETERMINATE (offset)	Strong ↓ (both down)

Key insight — fiscal and monetary push interest rates OPPOSITE ways

Expansionary FISCAL: gov borrows → DLF right → real rate ↑ (CROWDING OUT).

Expansionary MONETARY: Fed adds liquidity → MS right → real rate ↓.

When the Fed 'accommodates' expansionary fiscal: cancels out crowding out, but at the cost of more inflation in the long run.

This explains why politicians and central banks can disagree — fiscal stimulus alone raises rates; the Fed can offset that but creates inflationary pressure.

7.2 — The Phillips Curve

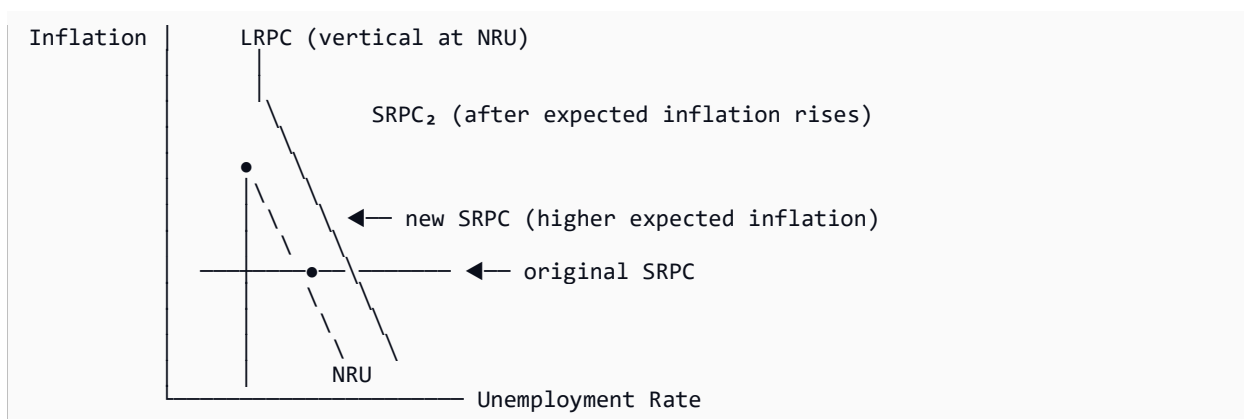
Phillips curve = inflation (Y) vs unemployment (X). It's the AD/AS model translated to the inflation-unemployment plane.

Short-Run Phillips Curve (SRPC) — downward sloping

In the SR there's a TRADE-OFF: lower unemployment costs more inflation. SRPC corresponds to AD shifts moving along the SRAS curve.

Long-Run Phillips Curve (LRPC) — vertical at NRU

In the LR, NO TRADE-OFF. The economy returns to NRU regardless of inflation. LRPC is vertical at the natural rate.



How AD/AS events map to Phillips curve

AD/AS Event	Movement on SRPC	Shift of SRPC	Shift of LRPC
AD shifts (demand-side change)	MOVES along	No	No
SRAS shifts (supply shock)	—	Shifts (left if positive shock, right if negative)	No
LRAS shifts (long-run growth, NRU change)	—	—	Shifts (left if NRU falls)
Expected inflation rises	—	Shifts RIGHT	No

Critical translation rules

AD right → MOVE up-left on SRPC. NO shift.

AD left → MOVE down-right on SRPC. NO shift.

SRAS left (cost-push, oil shock) → SRPC SHIFTS RIGHT. Trade-off worsens.

SRAS right (positive supply shock) → SRPC SHIFTS LEFT. Trade-off improves.

LRAS right (long-run growth) → LRPC may shift LEFT (NRU falls if labor markets improve).

Expected inflation up → SRPC SHIFTS RIGHT.

READING THE PHILLIPS GRAPH (your Q14)

For a given inflation rate, trace horizontally to the SRPC, drop down to read U.

For a given U, trace up to SRPC, read inflation off Y-axis.

LRPC is the vertical line — represents long-run U at NRU.

Trap: don't use LRPC unless the question explicitly asks about long-run.

Trap — 'Is there a long-run trade-off?'

NO. Repeatedly stimulating to push U below NRU only causes accelerating inflation — the SRPC keeps shifting right as expectations adjust. Eventually U returns to NRU at a higher inflation rate. If asked: 'Can the Fed permanently reduce unemployment?' → NO.

7.3 — Money Growth and Inflation (Quantity Theory)

Quantity Equation: $M \times V = P \times Q$

M = Money supply, V = Velocity, P = Price level, Q = Real output

Quantity Theory: V is roughly stable, Q grows slowly, so changes in M translate mostly into changes in P.

% change in M $\approx$ % change in P + % change in Q (when V constant)

Worked example

Money supply grows 8%/year. Real output grows 2%/year. Velocity constant.

Then inflation $\approx 8\% - 2\% = 6\%$.

Country with 30%/year money growth (assuming 2% real growth) → roughly 28% inflation.

Hyperinflation cases (Zimbabwe, Weimar Germany).

Long-run neutrality of money

In the LONG RUN, money is NEUTRAL — changes in MS only affect prices, not real output.

In the short run, money matters because of sticky wages/prices (the Keynesian story).

This is why central banks can't permanently lower unemployment — they can only affect short-run cyclical fluctuations.

7.4 — Government Deficits and the National Debt

Budget deficit = gov spending > tax revenue in a year. **National debt** = cumulative sum of past deficits.

Cyclical vs structural deficit

- **Cyclical deficit:** caused by the business cycle. Recession → less tax revenue, more transfers → deficit even with same policies. Self-corrects in expansion.
- **Structural deficit:** the part that would exist EVEN AT FULL EMPLOYMENT. Driven by spending and tax policies. Long-term problem.

When does a deficit matter?

Cyclical deficit during a recession: NORMAL and self-correcting.

Structural deficit at full employment: PROBLEM. Crowds out private investment, reduces long-run growth.

Persistent deficits → growing debt → growing interest payments → less fiscal flexibility.

7.5 — Crowding Out (high-frequency FRQ topic)

Mechanism: government borrowing increases demand for loanable funds, raising real interest rate, which reduces private investment.

The chain — graphically

1. Government runs deficit. To finance it, gov BORROWS.
2. DLF shifts RIGHT in loanable funds market.
3. Real interest rate ↑.
4. Higher real rate → private firms invest less. Households borrow less.
5. Private investment is CROWDED OUT.
6. Lower private investment → smaller capital stock → LRAS grows less than it would have. Long-run growth is dampened.

This is why structural deficits hurt long-run growth even when they don't immediately cause inflation.

(Your Q15 used this exact chain to determine dollar appreciation: deficit → DLF right → real rate up → capital inflow → dollar up.)

When does crowding out NOT happen?

If the Fed simultaneously runs expansionary monetary policy ('accommodates' the fiscal expansion), it adds reserves to the system, shifts SLF right, and the real rate doesn't rise → no crowding out.

But: this comes at the cost of higher inflation in the long run.

7.6 — Economic Growth

Growth = LRAS shifts RIGHT (potential output rises). Equivalent to PPC shifting outward. (Your Q13 tested this directly.)

Sources of growth

- **Capital formation:** investment in physical capital (factories, machines) AND human capital (education, skills). REQUIRES SAVINGS.
- **Technological progress:** better methods → more output per unit input.
- **Labor force expansion:** immigration, higher LFPR, population growth.
- **Resource discovery / quality.**
- **Productivity gains** from infrastructure, organization, education.

Why savings drives growth — the loanable funds connection

If everyone consumes everything, there's nothing left to invest in capital.

Higher savings → SLF shifts right → lower real rate → higher I → larger capital stock → LRAS shifts right → potential GDP grows.

This is THE central long-run insight.

(Your Q12 — business tax credit — tested this: incentive raises I → capital stock → real output.)

Effects of growth

- LRAS shifts RIGHT.
- Eventually SRAS also shifts right (more capacity over time).
- PL falls (long run); RGDP rises.
- PPC shifts outward.
- LRPC may shift left (if NRU falls due to better labor markets).

7.7 — Public Policy and Growth (supply-side)

Policy	How it affects growth	Trade-offs
Investment tax credits	↑ capital formation	Reduces gov revenue / risk of deficit
Lower marginal tax rates	Incentivizes work + investment	May reduce gov revenue
Infrastructure spending	Lowers production costs, ↑ productivity	Crowding out concern
Education subsidies	↑ human capital	Long lag before payoff
R&D tax credits	↑ technology	Cost
Free trade	Larger labor pool, comparative advantage	Distributional concerns
Deregulation	Lower costs, more entry	Possible loss of consumer/environmental protection

Supply-side fiscal vs demand-side fiscal

Demand-side: tax cuts/spending boost AD. Short-run focus.

Supply-side: tax cuts/credits encourage long-run capital formation, R&D, labor supply. LRAS shifts right.

Some policies (broad tax cuts, investment tax credits) shift BOTH AD and LRAS right.

Pure supply-side policies (R&D credits, education subsidies) primarily shift LRAS without much immediate AD effect.

7.8 — Unit 5 Traps

Trap 1 — Confusing growth with recovery

Moving INTO the PPC from inside (recession recovery) $\neq$ growth. That's just closing a recessionary gap.

Growth = the ENTIRE PPC shifts outward. The LRAS also shifts.

FRQ language: 'long-run growth' or 'potential output increase' $\rightarrow$ LRAS shift.

Trap 2 — 'Long-run effect of expansionary monetary policy on real output?'

ZERO. Long-run neutrality of money.

It only changes the price level in the long run. Real output returns to potential.

Many students answer 'real GDP rises' — wrong in the long run.

Trap 3 — Crowding out direction

Government deficit $\uparrow \rightarrow$ DLF SHIFTS RIGHT (not left — gov is borrowing MORE).

Real rate rises. Private investment falls.

Trap 4 — Phillips curve direction

Demand-side AD shift: MOVES along SRPC. NO shift.

Supply-side SRAS shift: SRPC shifts.

Long-run growth (NRU change): LRPC shifts.

Read the question: identify which type of shock first, then decide whether SRPC moves or shifts.

Part 8 — Units 1, 2, 6 Compressed Reference

Combined exam weight: ~30%. Lower-priority units. Compressed reference covering everything you need.

8.1 — Unit 1 quick reference (5–10% weight)

Comparative advantage

OUTPUT vs INPUT — quick rules

OUTPUT TABLE (units of good per worker): bigger number = absolute advantage. $OC = OTHER / OWN$.

INPUT TABLE (workers per unit of good): smaller number = absolute advantage. $OC = OWN / OTHER$.

Country with LOWER OC has comparative advantage in that good.

Trade benefits both when terms of trade fall BETWEEN the two countries' OCs.

Demand and supply shifters

Demand (TRIBE): Tastes, Related goods, Income, Buyers, Expectations.

Supply: Resource/input prices, Productivity/tech, Number of sellers, Taxes/Subsidies, Expectations (opposite direction from demand — expect price $\uparrow \rightarrow S$ left).

Market disequilibrium

- **Price ceiling** (binding when below equilibrium): causes SHORTAGE.
- **Price floor** (binding when above equilibrium): causes SURPLUS.

8.2 — Unit 2 quick reference (12–17% weight)

Circular Flow

- **Product market:** firms sell goods, households buy. Money flows clockwise from households as spending.
- **Factor market:** households sell labor/capital/land, firms pay wages/interest/rent.
- **Government:** collects taxes, makes purchases, transfers payments.
- **Foreign sector:** exports money in, imports money out. $NX = X - M$.

GDP — what counts, what doesn't

$$GDP = C + I + G + (X - M)$$

COUNTED: final goods, NEW housing (I), inventory changes (I) — your Q3 answer, gov wages (G), exports (X).

NOT COUNTED: used goods, intermediate goods, financial transactions (stocks/bonds), transfer payments, household production, illegal economy, nonmarket activities.

Unemployment formulas

Labor Force = Employed + Unemployed (actively seeking)

Unemployment Rate = Unemployed / LF × 100

LFPR = LF / Working-Age Population × 100

NRU = Frictional + Structural

Discouraged worker trick

Discouraged workers are NOT in the labor force.

When they leave, official rate FALLS even though true joblessness didn't improve.

When they re-enter, official rate RISES even though true joblessness didn't worsen.

Always remember: official rate UNDERSTATES true unemployment when discouraged workers exist.

Types of unemployment

- **Frictional:** voluntary, between jobs, temporary.
- **Structural:** skills mismatch, industry shift.
- **Cyclical:** recession-driven. The ONLY type tied to the business cycle.
- **Seasonal:** predictable cycle (lifeguards, ski instructors).

Price indices

CPI = (Cost of basket now / Cost of basket in base year) × 100

Inflation = (CPI₂ - CPI₁) / CPI₁ × 100

GDP Deflator = (Nominal GDP / Real GDP) × 100

Real GDP = (Nominal GDP / GDP Deflator) × 100

CPI vs GDP Deflator

CPI = consumer basket. INCLUDES imports.

GDP Deflator = all domestic production. EXCLUDES imports.

Imported oil spike: CPI rises a lot, GDP Deflator barely moves.

Domestic steel rises: GDP Deflator rises, CPI may not (if not in consumer basket directly).

Inflation winners and losers

Memorize this list

WINNERS from UNEXPECTED inflation:

- Borrowers (pay back with cheaper \$).
- Government with debt.
- Asset/property owners.

LOSERS:

- Lenders/creditors (paid back with cheaper \$).
- People on fixed income (retirees on fixed pension).
- Workers with fixed nominal wages.
- Cash holders / savers in low-interest accounts.

ONLY surprise inflation redistributes. Expected inflation is priced into nominal rates.

Demand-pull vs Cost-push

- **Demand-pull:** AD shifts right. PL $\uparrow$, RGDP $\uparrow$, U $\downarrow$. (Boom.)
- **Cost-push:** SRAS shifts left. PL $\uparrow$, RGDP $\downarrow$, U $\uparrow$. STAGFLATION.

8.3 — Unit 6 (Open Economy) quick reference (10–13% weight)**Balance of Payments**

$$\text{Current Account} + \text{Financial Account} = 0$$

- **Current Account:** trade in goods/services, income from abroad, transfers.
- **Financial Account:** FDI, portfolio investment, reserve transactions.
- **Implication:** a current account DEFICIT must be matched by a financial account SURPLUS. Trade deficits come with capital inflows.

Forex Market — drawing the graph

For the US dollar:

- Y-axis: exchange rate (e.g., yen/USD).
- X-axis: quantity of dollars.
- D_USD (downward): foreigners buying US goods/assets.
- S_USD (upward): Americans demanding foreign currency.

Currency direction reading

If Y-axis is YEN per DOLLAR (price of dollar in yen):

- Higher number = dollar APPRECIATES.
- D_USD shifts right $\rightarrow$ dollar appreciates.

If Y-axis is DOLLAR per YEN: opposite reading.
 READ THE AXIS LABEL FIRST. Always.

Forex shifters

Change	Effect on USD
US real interest rate $\uparrow$	Appreciates ($D_USD \uparrow$)
US real interest rate $\downarrow$	Depreciates
US inflation $\uparrow$	Depreciates (US goods less attractive)
US income $\uparrow$	Depreciates (more imports $\rightarrow S_USD \uparrow$)
Foreign income $\uparrow$	Appreciates (foreigners buy more US exports)
Foreign tastes for US goods $\uparrow$	Appreciates
US tastes for foreign goods $\uparrow$	Depreciates
Speculation USD will rise	Appreciates (self-fulfilling)

Effect of currency changes on net exports

Strong vs weak currency for trade

DOLLAR APPRECIATES (strong dollar):

- US goods MORE EXPENSIVE for foreigners $\rightarrow X \downarrow$.
- Foreign goods CHEAPER for Americans $\rightarrow M \uparrow$.
- Net exports $\downarrow \rightarrow AD$ shifts LEFT.
- (Your FRQ #2 Part F: zeet appreciated $\rightarrow$ exports decrease.)

DOLLAR DEPRECIATES (weak dollar):

- US goods CHEAPER for foreigners $\rightarrow X \uparrow$.
- Foreign goods MORE EXPENSIVE for Americans $\rightarrow M \downarrow$.
- Net exports $\uparrow \rightarrow AD$ shifts RIGHT.

Real interest rates and capital flows — THE critical chain

Higher US rates $\rightarrow$ currency appreciation chain (your Q15)

1. US real interest rate $\uparrow$ (relative to other countries).
2. Foreign investors want US bonds (higher returns).
3. They demand DOLLARS to buy those bonds.

4. D_USD shifts RIGHT. Dollar APPRECIATES.
 5. US exports more expensive abroad $\rightarrow X \downarrow$.
 6. Foreign goods cheaper for Americans $\rightarrow M \uparrow$.
 7. Net exports $\downarrow \rightarrow AD \downarrow$ (partially offsets the higher rate's domestic stimulus).
- Memorize this. Multiple FRQ questions hinge on it.

Relative price levels and exports (your FRQ #1 Part B)

How relative inflation affects exports

If country A's price level rises FASTER than its trading partners',

Country A's goods become RELATIVELY MORE EXPENSIVE for foreigners.

Foreign importers buy FEWER goods from country A.

Exports DECREASE.

Even WITHOUT a currency change, relative inflation can hurt exports.

(Fehran's case: 25% inflation, partners flat $\rightarrow$ EVs more expensive $\rightarrow$ fewer exports in 2012.)

Part 9 — Graph Reading Drills

Q1 (PPC) and Q14 (Phillips) in your practice tested graph-reading directly. The exam loves these. Master the technique.

9.1 — PPC Reading (Q1 type)

Technique

1. Find two known points on the curve. Calculate slope = rise/run = $\Delta Y / \Delta X$.
2. Write the line equation: $Y = \max_Y - |\text{slope}| \times X$.
3. For each candidate point (X_c , Y_c): plug X_c into the equation. Get $\max_Y$ at X_c .
4. Compare:
 - $Y_c < \max_Y$: ATTAINABLE (inside curve).
 - $Y_c = \max_Y$: ATTAINABLE (on curve).
 - $Y_c > \max_Y$: UNATTAINABLE (outside curve).

Total elapsed time: ~30 seconds.

Drill 1

PPC for Country A: produces shoes and bread. Endpoints (0 shoes, 100 bread) and (50 shoes, 0 bread). Linear PPC.

Slope = $-100/50 = -2$ bread per shoe.

Equation: bread = $100 - 2 \times \text{shoes}$.

Test: 20 shoes, 60 bread. Max bread at 20 shoes = $100 - 40 = 60$. Exactly 60. ATTAINABLE (on curve).

Test: 25 shoes, 60 bread. Max at 25 shoes = 50. Need $60 > 50$. UNATTAINABLE.

Test: 10 shoes, 70 bread. Max at 10 shoes = 80. $Y_c = 70 < 80$. ATTAINABLE (inside).

Drill 2 (mirrors your Q1 setup)

PPC: (0, 12) to (30, 0). Slope = $-12/30 = -0.4$. bread = $12 - 0.4 \times \text{shoes}$.

Practice: which is unattainable?

- a) 10 shoes, 8 bread $\rightarrow \max = 12 - 4 = 8$. ON curve. Attainable.
- b) 20 shoes, 5 bread $\rightarrow \max = 12 - 8 = 4$. $Y_c = 5 > 4$. UNATTAINABLE. $\leftarrow$
- c) 15 shoes, 5 bread $\rightarrow \max = 12 - 6 = 6$. Inside. Attainable.

Special case — bowed-out PPC

Nonlinear PPCs (bowed outward) reflect INCREASING opportunity cost.

You can't compute slope from two endpoints — slope changes along the curve.

For these, judge visually: is the candidate point clearly inside, on, or outside the visible curve? AP graphs always make this readable.

9.2 — Phillips Curve Reading (Q14 type)

Technique

1. SRPC slopes downward. Identify two known points to determine the line.
2. For 'given inflation, find U': trace HORIZONTALLY from inflation value to the SRPC, then drop DOWN to read U on x-axis.
3. For 'given U, find inflation': trace UP from U to the SRPC, then read inflation off y-axis.
4. The LRPC is vertical at NRU. If question says 'long run,' use the LRPC.

Drill — your Q14 worked through

SRPC visible points: (1, 8), (2, 6), (3, 4), (4, 2), (5, 1). LRPC vertical at U=4.

Question: at inflation = 6%, what's U?

Trace from inflation = 6 horizontally to SRPC → hits the curve at the (2, 6) point. So U = 2%.

Answer: B (2%).

Practice — Phillips reading

Drill A: SRPC passes through (1, 8) and (5, 0). Slope = $(0-8)/(5-1) = -2$ inflation per U. Equation: inflation = $10 - 2U$.

At inflation = 4%, U = ? → $4 = 10 - 2U \rightarrow U = 3\%$.

At U = 1%, inflation = ? → $10 - 2 = 8\%$.

Drill B: LRPC vertical at NRU = 5. Inflation can be ANY value — U stays at 5% in long run.

Drill C: SRPC shifts RIGHT. Original (3, 4). New (4, 4). The trade-off worsened: same inflation, higher U.

Drill D: AD shifts right (demand-pull). Movement is ALONG SRPC, up-and-left. From (3, 4) → (2, 6).

9.3 — AD/AS Graph Reading

Identifying the gap from a graph

Find LRAS (vertical line). Find AD/SRAS intersection.

If intersection is LEFT of LRAS: RECESSIONARY GAP ($RGDP < \text{potential}$).

If intersection is ON LRAS: LONG-RUN EQUILIBRIUM.

If intersection is RIGHT of LRAS: INFLATIONARY GAP ($RGDP > \text{potential}$).

Worked through — your Q7

Graph shows equilibrium at H. Question asks how to REDUCE PL in short run.

(Whether H is in inflationary, recessionary, or full employment doesn't matter — we need contractionary policy regardless.)

Want AD shift LEFT.

Increasing IORB (interest on reserves) → contractionary monetary → AD LEFT → PL DOWN. ✓

Answer: C.

9.4 — Speed Drill: Identify the Graph in <5 seconds

For each axis pair, name the model:

1. Y: Price Level. X: Real GDP. → AD/AS
2. Y: Nominal interest rate. X: Quantity of money. → Money Market
3. Y: Real interest rate. X: Quantity of loanable funds. → Loanable Funds
4. Y: Inflation rate. X: Unemployment rate. → Phillips Curve
5. Y: Exchange rate. X: Quantity of currency. → Forex
6. Y: Federal Funds Rate. X: Quantity of reserves. → Reserve Market

Part 10 — Master Chain Library

Every 'trace through all markets' FRQ reduces to one of these 12 chains. Memorize them.

10.1 — Why chains matter

How AP Macro tests reasoning

Long FRQs ask you to trace effects through 3–5 graphs in sequence (your FRQ #2 has six parts, each a step in a chain).

Students who memorize concepts but don't practice chains lose points by forgetting steps or doing them out of order.

Memorize the 12 chains. On the exam, identify which chain applies, then write it out.

Each chain ends at AD/AS with effects on PL, RGDP, U. Phillips and forex are auxiliary.

10.2 — The 12 Chains

Chain 1 — EXPANSIONARY FISCAL POLICY ($\uparrow G$ or $\downarrow T$)

Trace

1. Government increases G (or cuts T).
2. AD shifts RIGHT directly. $\rightarrow PL \uparrow, RGDP \uparrow, U \downarrow$.
3. Government borrows more $\rightarrow DLF$ shifts RIGHT.
4. Real interest rate $\uparrow \rightarrow$ private $I \downarrow$ (CROWDING OUT).
5. Higher US rates $\rightarrow$ capital INFLOW $\rightarrow D_USD$ right $\rightarrow$ dollar APPRECIATES $\rightarrow X \downarrow, M \uparrow \rightarrow AD \downarrow$ slightly.
6. Phillips: MOVE up-left along SRPC.

Chain 2 — CONTRACTIONARY FISCAL POLICY ($\downarrow G$ or $\uparrow T$)

Trace (mirror of Chain 1)

1. $\downarrow G$ or $\uparrow T \rightarrow AD$ shifts LEFT $\rightarrow PL \downarrow, RGDP \downarrow, U \uparrow$.
2. Less gov borrowing $\rightarrow DLF$ LEFT $\rightarrow$ real rate $\downarrow \rightarrow I \uparrow$ slightly.
3. Lower US rates $\rightarrow$ capital outflow $\rightarrow S_USD$ right $\rightarrow$ dollar DEPRECIATES $\rightarrow X \uparrow, M \downarrow \rightarrow AD \uparrow$ slightly.
4. Phillips: MOVE down-right along SRPC.

Chain 3 — EXPANSIONARY MONETARY POLICY (Fed buys bonds, lowers IORB)

Trace

1. Fed BUYS bonds → bond prices ↑ → interest rates ↓.
2. Money market: MS shifts RIGHT → nominal i ↓.
3. Loanable funds: SLF shifts RIGHT → real rate ↓.
4. Lower real rate → I ↑, durables- C ↑ → AD shifts RIGHT.
5. Forex: lower US rates → capital outflow → dollar DEPRECIATES → X ↑, M ↓ → AD ↑ MORE.
6. AD/AS: PL ↑, RGDP ↑, U ↓.
7. Phillips: MOVE up-left along SRPC.

Chain 4 — CONTRACTIONARY MONETARY POLICY (Fed sells bonds, raises IORB)**Trace — your FRQ #2 chain**

1. Fed SELLS bonds (or raises IORB) → bond prices ↓ → rates ↑.
 2. MS shifts LEFT → nominal i ↑.
 3. Real rate ↑ → I ↓, durables- C ↓ → AD shifts LEFT.
 4. Forex: higher US rates → capital INFLOW → D_{USD} right → dollar APPRECIATES → X ↓, M ↑ → AD ↓ MORE.
 5. PL ↓, RGDP ↓, U ↑.
 6. Phillips: MOVE down-right along SRPC.
- (This is exactly the chain underlying your FRQ #2 about Zeetoland.)

Chain 5 — POSITIVE SUPPLY SHOCK (productivity gain, oil price drop)**Trace**

1. SRAS shifts RIGHT.
 2. AD/AS: PL ↓, RGDP ↑, U ↓ — the Goldilocks scenario.
 3. Phillips curve: SRPC SHIFTS LEFT (lower inflation possible at every U level).
 4. If permanent, also LRAS RIGHT (productivity is a permanent improvement).
- (Your Q11 tested this — only positive supply-side shock satisfies $RGDP \uparrow$ SR + PL ↓ LR.)

Chain 6 — NEGATIVE SUPPLY SHOCK (oil spike, war, pandemic disruption)**Trace — STAGFLATION**

1. SRAS shifts LEFT.
2. PL ↑ AND RGDP ↓ AND U ↑ — STAGFLATION.
3. Phillips curve: SRPC SHIFTS RIGHT.

4. Policy bind: easing AD lowers U but worsens inflation; tightening AD lowers inflation but worsens U.
5. If permanent (resource depletion), LRAS also shifts LEFT.

Chain 7 — LONG-RUN ECONOMIC GROWTH (capital, tech, labor expansion)

Trace

1. More capital, technology, or labor → LRAS shifts RIGHT (potential GDP rises). PPC shifts outward (your Q13).
2. SRAS also eventually shifts right (more capacity).
3. Long-run: PL ↓, RGDP ↑.
4. LRPC may shift LEFT if NRU falls (improved labor markets).
5. AD doesn't necessarily shift — but if income rises, consumption rises, AD shifts right too.

Chain 8 — INCREASE IN FOREIGN INCOME (trade-partner growth)

Trace

1. Foreign income ↑ → foreigners buy MORE US exports.
2. D_USD shifts RIGHT → dollar APPRECIATES.
3. Net exports ↑ (X up) → AD shifts RIGHT.
4. AD/AS: PL ↑, RGDP ↑, U ↓.
5. Phillips: MOVE up-left along SRPC.
6. Current account improves (more exports).

Chain 9 — DOMESTIC RECESSION (consumer/business confidence collapse)

Trace

1. C and/or I fall → AD shifts LEFT.
2. PL ↓, RGDP ↓, U ↑ (cyclical U emerges).
3. Phillips: MOVE down-right along SRPC.
4. Lower income → MD shifts left → nominal rate ↓.
5. Less borrowing demand → DLF left → real rate ↓.
6. Lower US rates → capital outflow → S_USD right → dollar DEPRECIATES → X ↑, M ↓ → AD ↑ slightly (auto-stabilizer).
7. Self-correction: high U eventually → wages fall → SRAS RIGHT → return to LRAS at lower PL.

Chain 10 — INFLATIONARY BOOM (consumer/business confidence surge)

Trace

1. C and/or I rise → AD shifts RIGHT.
2. PL ↑, RGDP ↑ (above potential), U ↓ (below NRU — your FRQ #2 setup).
3. Phillips: MOVE up-left along SRPC.
4. Higher income → MD ↑ → nominal rate ↑.
5. Higher demand for borrowing → DLF right → real rate ↑.
6. Higher US rates → capital INFLOW → D_USD right → dollar APPRECIATES → X ↓, M ↑ → AD ↓ slightly.
7. Self-correction: tight labor → wages rise → SRAS LEFT → return to LRAS at higher PL.

Chain 11 — INCREASE IN HOUSEHOLD SAVINGS**Trace**

1. People save more, consume less → SLF shifts RIGHT in loanable funds.
2. Real interest rate ↓.
3. Lower r → investment ↑ → AD shifts RIGHT slightly.
4. BUT C ↓ pulls AD LEFT initially.
5. Net short-run effect on AD: ambiguous (C falling vs I rising).
6. Long-run: more I → more capital formation → LRAS shifts RIGHT → long-run growth.
(The 'savings paradox' — short-run AD may fall, but long-run growth improves.)

Chain 12 — INCREASE IN MONEY SUPPLY GROWTH RATE (long-run)**Trace (long-run)**

1. Higher M growth → quantity theory: $\% \Delta P \approx \% \Delta M - \% \Delta Q$ → inflation rises.
2. Expected inflation ↑ → SRPC SHIFTS RIGHT (worse trade-off).
3. Workers demand higher wages → SRAS shifts left.
4. Self-correction returns RGDP to potential, but at higher PL.
5. Long-run neutrality of money: real GDP unchanged, prices higher.

10.3 — Effect Lookup Table

Shock / Policy	PL	RGDP	U	Real i	Nominal i	Currency	Phillips
Expansionary fiscal	↑	↑	↓	↑	↑	Appreciate	Up-left SRPC

Shock / Policy	PL	RGDP	U	Real i	Nominal i	Currency	Phillips
Contractionary fiscal	↓	↓	↑	↓	↓	Depreciate	Down-right SRPC
Expansionary monetary	↑	↑	↓	↓	↓	Depreciate	Up-left SRPC
Contractionary monetary	↓	↓	↑	↑	↑	Appreciate	Down-right SRPC
Positive AD shock	↑	↑	↓	↑	↑	Appreciate	Up-left SRPC
Negative AD shock	↓	↓	↑	↓	↓	Depreciate	Down-right SRPC
Positive supply shock	↓	↑	↓	↓	↓	Depreciate	SRPC LEFT
Negative supply (stagflation)	↑	↓	↑	↑	↑	Mixed	SRPC RIGHT
LR growth (LRAS R)	↓ LR	↑	↓ NRU	?	?	?	LRPC LEFT
Foreign income ↑	↑	↑	↓	?	?	Appreciate	Up-left SRPC

Part 11 — Scenario Drills

30 scenarios. Cover the answer first; reason chain; then check. Target: 2 minutes per scenario. Builds chain-reasoning muscle memory under simulated FRQ time pressure.

Scenario 1

Q: The Federal Reserve buys \$50 billion in government bonds. Trace the effects on the money market, loanable funds market, AD/AS, and the value of the dollar.

Solution

Money market: MS shifts RIGHT → nominal interest rate FALLS.

Loanable funds: SLF shifts RIGHT → real interest rate FALLS.

Lower real rate → investment ↑, durables-C ↑ → AD shifts RIGHT → PL ↑, RGDP ↑, U ↓.

Forex: lower US rates → capital outflow → S_{USD} shifts right → dollar DEPRECIATES → X ↑, M ↓ → AD shifts RIGHT MORE.

Phillips: move up-left along SRPC.

Scenario 2

Q: Congress passes a \$200 billion tax cut. The MPC is 0.8. Calculate the maximum increase in real GDP, then identify the effect on real interest rates.

Solution

Tax multiplier = $-0.8 / 0.2 = -4$.

Maximum $\Delta GDP = \$200B \times 4 = \$800B$.

Real rate: gov revenue falls → deficit ↑ → gov borrows more → DLF shifts RIGHT → real rate RISES (CROWDING OUT).

Scenario 3

Q: An oil shock causes input prices to spike. Trace effects on AD/AS, the Phillips curve, and discuss what happens to real GDP and inflation.

Solution

SRAS shifts LEFT.

AD/AS: PL ↑ AND RGDP ↓ AND U ↑ — STAGFLATION.

Phillips: SRPC SHIFTS RIGHT (worse trade-off).

Policy dilemma: easing AD lowers U but worsens inflation; tightening AD lowers inflation but worsens U.

Scenario 4

Q: Foreign investors lose confidence and pull capital out of the US. Trace effects on the forex market, loanable funds, AD/AS, and trade balance.

Solution

Capital outflow → SLF shifts LEFT (less foreign capital available).

Forex: foreigners selling USD assets → S_{USD} right (or D_{USD} left) → dollar DEPRECIATES.

Loanable funds: SLF left → real rate UP.

Higher real rate → $I \downarrow$ → AD shifts LEFT.

BUT depreciated dollar → $X \uparrow$, $M \downarrow$ → AD shifts RIGHT (offsetting effect).

Net AD: depends on relative magnitudes. Trade balance improves.

Scenario 5

Q: Government raises both spending and taxes by \$300 billion (balanced budget). What happens to real GDP?

Solution

Balanced budget multiplier = 1.

$\Delta GDP = +\$300B \times \text{Spending mult} + \$300B \times \text{Tax mult}.$

If $MPC = 0.75$: $\Delta GDP = 300 \times 4 + 300 \times (-3) = 1,200 - 900 = \$300B.$

Real GDP rises by exactly the spending change.

Scenario 6

Q: Discouraged workers re-enter the labor force seeking jobs but don't find them. What happens to the official unemployment rate?

Solution

Official unemployment rate RISES.

Numerator (unemployed) and denominator (LF) both rise, but numerator rises more proportionally.

Labor market is NOT actually worsening — these workers were always jobless.

Shows the official rate UNDERSTATES true unemployment when discouraged workers are excluded.

Scenario 7

Q: A central bank doubles the money supply over 2 years; real output grows 2% per year. Estimate inflation rate.

Solution

$MV = PQ$. Assume V constant.

$\% \Delta M \approx \% \Delta P + \% \Delta Q$.

Money doubled in 2 years $\rightarrow \sim 100\%$ over 2 years.

$\Delta Q = 2\% \times 2 = 4\%$ over 2 years.

Inflation over 2 years $\approx 100\% - 4\% = 96\%$. Hyperinflation territory.

Scenario 8

Q: Recessionary gap. Fed lowers IORB. Trace through money market, loanable funds, AD/AS, and dollar. Will the gap close?

Solution

Lower IORB $\rightarrow$ reserve market: equilibrium rate falls $\rightarrow$ effectively expansionary.

Money market: MS effectively shifts RIGHT $\rightarrow$ nominal rate falls.

Loanable funds: SLF shifts RIGHT $\rightarrow$ real rate falls.

Lower real rate $\rightarrow I \uparrow \rightarrow$ AD shifts RIGHT.

Forex: lower US rates $\rightarrow$ capital outflow $\rightarrow$ dollar DEPRECIATES $\rightarrow X \uparrow, M \downarrow \rightarrow$ AD $\uparrow$ more.

AD/AS: PL $\uparrow$, RGDP $\uparrow$, U $\downarrow$.

If calibrated correctly, gap closes.

Scenario 9

Q: Country has current account deficit of \$300B. What does this imply about its financial account and capital flows?

Solution

Current Account + Financial Account = 0.

CA = $-\$300\text{B}$ $\rightarrow$ FA = $+\$300\text{B}$.

Net capital INFLOW of \$300B (foreigners buy more US assets than Americans buy foreign).

Trade deficit and capital inflow are TWO SIDES of the same coin.

Scenario 10

Q: Fed announces it expects to raise interest rates next year. What happens to expected inflation, the SRPC, and current AD?

Solution

Expected future tightening $\rightarrow$ expected inflation FALLS.

Lower expected inflation $\rightarrow$ SRPC SHIFTS LEFT (better trade-off).

SRAS shifts RIGHT slightly (lower wage demands).

Investors anticipate lower returns → some pullback in current investment → AD slightly LEFT.
 Net effect: PL pressure down, RGDP effect ambiguous.

Scenario 11

Q: Real GDP = \$9,200B. Potential GDP = \$10,000B. MPC = 0.6. What change in G closes the gap?

Solution

Output gap = $10,000 - 9,200 = \$800\text{B}$.

Spending mult = $1/(1-0.6) = 1/0.4 = 2.5$.

$\Delta G = \$800\text{B} / 2.5 = \320B INCREASE.

Scenario 12

Q: Bank has \$5,000 deposits. RRR = 25%. Total reserves = \$2,000. Find required, excess, money mult, max new money creation.

Solution

Required = $\$5,000 \times 0.25 = \$1,250$.

Excess = $\$2,000 - \$1,250 = \$750$.

Money multiplier = $1/0.25 = 4$.

Max new money = $\$750 \times 4 = \$3,000$.

Scenario 13

Q: Surge in consumer confidence. Trace effects on AD/AS, money market, loanable funds, and dollar.

Solution

C rises → AD shifts RIGHT → PL ↑, RGDP ↑, U ↓.

Higher income/PL → MD shifts RIGHT → nominal rate ↑.

More borrowing demand → DLF shifts RIGHT → real rate ↑.

Higher US rates → capital INFLOW → D_USD right → dollar APPRECIATES → X ↓, M ↑ → AD ↓ slightly.

Phillips: move up-left along SRPC.

Scenario 14

Q: Long-run equilibrium economy experiences a productivity breakthrough. Short-run and long-run effects?

Solution

Short-run: SRAS shifts RIGHT (lower per-unit costs).
 Long-run: LRAS shifts RIGHT (potential output rises). PPC outward.
 AD/AS: PL ↓, RGDP ↑, U ↓.
 Phillips: SRPC LEFT. LRPC may shift LEFT if NRU falls.
 Real rate: depends on whether savings or investment demand rises more.

Scenario 15

Q: Government cuts personal income taxes by \$100B. MPC = 0.75. (a) Calculate ΔAD . (b) Effect on loanable funds. (c) Effect on dollar.

Solution

- (a) Tax mult = $-0.75/0.25 = -3$. $\Delta AD \approx \$100B \times 3 = \$300B$ (AD shifts right).
 (b) Tax cut → larger deficit → DLF shifts RIGHT → real rate ↑.
 (c) Higher US rates → capital INFLOW → D_USD right → dollar APPRECIATES → X ↓, M ↑.

Scenario 16

Q: Economy at full employment. Congress passes large expansionary fiscal. Short run vs long run effects?

Solution

Short run: AD shifts RIGHT → PL ↑, RGDP ↑ above potential, U ↓ below NRU. Inflationary gap.
 Long run: tight labor market → wages rise → SRAS shifts LEFT → returns to LRAS at HIGHER PL.
 Long-run real GDP unchanged. Only inflation, no real output gain.

Scenario 17

Q: Country: Population = 300M. Working-age population = 240M. Employed = 150M. Unemployed and seeking = 12M. Find LF, U rate, LFPR.

Solution

Labor Force = $150 + 12 = 162M$.
 U Rate = $12/162 \times 100 \approx 7.41\%$.
 LFPR = $162/240 \times 100 = 67.5\%$.

Scenario 18

Q: NRU = 5%. Current U = 8%. What type of unemployment exists and what gap is the economy in?

Solution

Cyclical = $8 - 5 = 3\%$. Frictional + Structural still exist (= 5%).

Cyclical $> 0 \rightarrow$ economy in a RECESSIONARY GAP. $RGDP < \text{potential}$.

Scenario 19

Q: CPI rose from 220 (2022) to 242 (2023). Calculate inflation. Worker's nominal wage rose \$30/hr to \$32/hr — what happened to real wage?

Solution

Inflation = $(242 - 220)/220 \times 100 = 10\%$.

Nominal wage growth = $(32 - 30)/30 \times 100 = 6.67\%$.

Approximate real wage change $\approx 6.67\% - 10\% = -3.33\%$. Real wage FELL.

Scenario 20

Q: Country runs expansionary fiscal AND expansionary monetary simultaneously. Real interest rate effect?

Solution

Fiscal alone: deficit $\rightarrow$ DLF right $\rightarrow$ real rate UP.

Monetary alone: Fed adds liquidity $\rightarrow$ SLF right $\rightarrow$ real rate DOWN.

Combined: forces OFFSET. Real rate change is INDETERMINATE.

AD shifts RIGHT strongly (both reinforce on AD).

If fiscal effect dominates, real rate rises (some crowding out).

If monetary fully accommodates, real rate barely changes — but inflation pressure builds long-run.

Scenario 21

Q: Dollar appreciates significantly. (a) Effect on net exports? (b) Effect on AD? (c) Effect on current account? (d) Effect on Phillips curve?

Solution

(a) NX falls (X down, M up).

(b) AD shifts LEFT.

(c) Current account worsens.

(d) AD left $\rightarrow$ MOVE down-right along SRPC. NO shift of SRPC itself.

Scenario 22

Q: Foreign demand for US goods surges. Fed wants to maintain price stability. What policy and why?

Solution

Foreign demand surge $\rightarrow X \uparrow \rightarrow AD$ shifts RIGHT $\rightarrow PL \uparrow$.

To prevent inflation, Fed should adopt CONTRACTIONARY monetary policy.

Effect: MS shifts left $\rightarrow$ nominal rate up $\rightarrow$ real rate up $\rightarrow I \downarrow \rightarrow AD$ shifts LEFT, offsetting.

Side effect: higher rates $\rightarrow$ capital inflow $\rightarrow$ dollar appreciates further $\rightarrow$ reinforces AD-left through net exports.

Scenario 23

Q: Permanent productivity increase from technology adoption. Long-run Phillips curve effect?

Solution

Productivity gains $\rightarrow$ SRAS RIGHT permanently $\rightarrow$ LRAS RIGHT (long-run).

Lower per-unit production costs $\rightarrow$ lower equilibrium PL at every U.

SRPC SHIFTS LEFT.

LRPC may shift LEFT (if NRU falls due to better labor market dynamics).

Scenario 24

Q: \$500B deficit financed entirely by new bond issuance. Effect on loanable funds, real rate, private investment, long-run growth?

Solution

DLF shifts RIGHT.

Real rate RISES.

Private investment FALLS (CROWDING OUT).

If gov uses funds for productive investment (infrastructure, education): LRAS may rise too — partially offsetting.

If gov uses for current consumption: LRAS unchanged $\rightarrow$ only crowding out $\rightarrow$ LOWER long-run growth.

Scenario 25

Q: Economy faces stagflation. Why is traditional fiscal/monetary policy ineffective?

Solution

Stagflation = $PL \uparrow$ AND $RGDP \downarrow$ AND $U \uparrow$ (caused by SRAS LEFT).

EXPANSIONARY policy: AD right $\rightarrow$ reduces U BUT worsens inflation. Bad.

CONTRACTIONARY policy: AD left $\rightarrow$ reduces inflation BUT worsens U. Bad.

Demand-side tools can't fix supply-side problem. Best response: SUPPLY-SIDE policies (productivity gains, alternatives, supply chain solutions) — shift SRAS back right.
This is why 1970s stagflation took ~10 years to resolve.

Scenario 26

Q: Country had Real GDP \$400B with population 40M. Five years later: Real GDP \$500B, population 50M. What happened to standard of living? Test the per-capita reasoning that the Fehran FRQ relied on.

Solution

Real GDP per capita Year 1 = $400/40 = \$10,000$.

Real GDP per capita Year 5 = $500/50 = \$10,000$.

Standard of living UNCHANGED.

Real GDP grew 25%, population grew 25% → per capita flat.

Standard of living tracks REAL GDP PER CAPITA, not raw RGDP. (Same logic as Fehran 2011→2012.)

Scenario 27

Q: Two countries trade. Country A's price level rises 30%. Country B's price level stays flat. No currency change. What happens to Country A's exports to B?

Solution

Country A's goods become 30% more expensive in relative terms.

Foreign importers (Country B) substitute toward cheaper alternatives.

A's exports DECREASE.

Even WITHOUT a currency change, relative inflation alone hurts exports.

(This is the logic underlying your FRQ #1 Part B about Fehran's electric vehicles.)

Scenario 28

Q: An economy is in short-run equilibrium with actual U = 4% and NRU = 6%. Identify the gap. Show what happens with NO policy intervention.

Solution

Actual U (4%) < NRU (6%) → INFLATIONARY GAP.

RGDP > potential GDP. AD/SRAS intersect RIGHT of LRAS.

Self-correction (no policy): tight labor market → wages rise → input costs rise → SRAS shifts LEFT.

Process continues until SRAS intersects AD on LRAS — back at potential.

Result: PL ↑, RGDP returns to potential, U returns to NRU.

(This is the Zeetoland setup in your FRQ #2 — but THERE, actual $U = 5\% < NRU = 6\%$ → still inflationary gap.)

Scenario 29

Q: Fed engages in contractionary monetary policy. Trace effects on (a) bond prices, (b) real output in short run, (c) net financial capital flows.

Solution

(a) Fed sells bonds (or raises IORB, signaling tighter policy). Bond supply rises → BOND PRICES FALL. (Inverse rate-price relationship.)

(b) MS shifts left → nominal rate up → real rate up → I and durables-C down → AD shifts LEFT → REAL OUTPUT FALLS in short run.

(c) Higher domestic real rates → foreign investors want our bonds → capital INFLOW (positive net financial capital flow into the country).

(All three are exact answers to your FRQ #2 Part D.)

Scenario 30

Q: Country has Nominal GDP \$1,500B in 2024 with deflator = 120. In 2023: Nominal \$1,400B with deflator = 110. Calculate (a) real GDP for both years, (b) real GDP growth rate, (c) inflation rate, (d) nominal GDP growth rate.

Solution

(a) Real GDP 2023 = $(1,400/110) \times 100 = \$1,272.7\text{B}$. Real GDP 2024 = $(1,500/120) \times 100 = \$1,250\text{B}$.

(b) Real GDP growth = $(1,250 - 1,272.7)/1,272.7 \times 100 \approx -1.78\%$ (real GDP FELL).

(c) Inflation = $(120 - 110)/110 \times 100 \approx 9.09\%$.

(d) Nominal growth = $(1,500 - 1,400)/1,400 \times 100 \approx 7.14\%$.

Cross-check: Nominal growth $\approx$ Real growth + Inflation → $7.14 \approx -1.78 + 9.09 = 7.31$. Close (slight rounding).

Part 12 — FRQ Mastery + Worked Solutions

FRQs are graded by RUBRIC. You either earn a point or you don't. Give the rubric exactly what it asks for, get full marks. This section trains the rubric mindset, then walks through both actual FRQs from your practice screenshots.

12.1 — Section II Structure (from your exam directions)

What you'll see on exam day

3 FRQs total. 1 hour total.

Suggested timing: 10 min READING period + 25 min long FRQ + 12 min each on FRQs 2 and 3.

Calculator allowed.

Pencil or black/dark blue pen.

Label parts (A, B, C) and sub-parts (i, ii, iii) clearly.

Correctly labeled graphs require: all axes labeled, all curves labeled, directional changes shown.

'Calculate' = MUST show how you arrived at the final answer.

Only work in the FREE-RESPONSE BOOKLET is scored. Scratch work is not graded.

12.2 — Decoding the FRQ Verbs

Verb	What's required	Pitfall
Identify	Just name it. One word/phrase suffices.	Don't over-explain.
Calculate	Show numerical work AND give final answer with units.	No work shown = lose the point even if answer is right.
Show on graph	Draw + label clearly with arrows or letters.	Unlabeled curves earn no credit.
Explain	Give the cause-and-effect chain.	Stating Y without 'because of X' loses the point.
What will happen to X	State direction (↑/↓/no change/uncertain).	Don't just describe — answer directly.
Will X increase, decrease, or stay the same? Explain.	1) State direction. 2) Give cause/effect.	Both pieces needed for full credit.

Key insight — answer EXACTLY what's asked

If asked for the effect on real GDP, answer with real GDP — not unemployment.

If asked to draw, you draw. If asked to explain, you write words.

Most lost points come from answering a related-but-different question.

Read every part TWICE before answering.

12.3 — Universal Graph Drawing Rules

Every graph on the AP exam needs:

1. **Title / model name** at the top ('AD/AS Model,' 'Money Market,' 'Loanable Funds Market,' etc.).
2. **Both axes labeled** with the correct variable name.
3. **All curves labeled** (D, S, AD, SRAS, LRAS, MD, MS, DLF, SLF, etc.).
4. **Equilibrium points marked** with a dot or letter.
5. **Dotted lines** from equilibrium to BOTH axes, showing equilibrium values labeled (P^* , Q^* , i^* , r^* , e^* , etc.).
6. **If a shift occurred:** show the original AND new curve, label both (e.g., AD_1 and AD_2), use arrows to show direction of shift.
7. **New equilibrium point** with new dotted lines to new axis values.

The 6-second graph check before moving on

1. Title at top? ✓
2. Axes labeled? ✓
3. Curves labeled? ✓
4. Equilibrium dotted lines to both axes? ✓
5. New curves labeled (if shifted)? ✓
6. Direction-of-change arrows visible? ✓

If any are missing, you've potentially lost a point. Fix it before moving on.

12.4 — The 10 Required Graphs

#	Graph	Y-axis	X-axis	Curves
1	PPC	Good A	Good B	Curve (linear or bowed-out)
2	Supply & Demand	Price	Quantity	D, S
3	Business Cycle	Real GDP	Time	Wave + trend line
4	Circular Flow	—	—	Households, Firms, Gov, markets

#	Graph	Y-axis	X-axis	Curves
5	AD/AS	Price Level	Real GDP	AD, SRAS, LRAS
6	Loanable Funds	Real Interest Rate	Q of Loanable Funds	D _L , S _L
7	Money Market	Nominal Interest Rate	Q of Money	MD, MS
8	Phillips Curve	Inflation Rate	Unemployment Rate	SRPC, LRPC
9	Reserve Market	Federal Funds Rate	Q of Reserves	RD (stepped), RS
10	Forex Market	Exchange Rate	Q of Currency	D _{currency} , S _{currency}

12.5 — Worked Solution: Actual FRQ #1 (Fehran)

FRQ #1 setup (from your practice)

Fehran macro statistics:

Year 2011: Nominal GDP \$150,000. Tax revenues \$25,000. Government outlays \$25,000.

Consumption \$75,000. Population 100. GDP Deflator 100.

Year 2012: Nominal GDP \$225,000. Tax revenues \$30,000. Government outlays \$35,000.

Consumption \$100,000. Population 120. GDP Deflator 125.

Question has 4 parts: A (calculate), B (relative price → exports reasoning), C (standard of living), D (deficit → debt).

Part A.i — Real GDP for 2012

Solution (show work — required)

$$\text{Real GDP} = (\text{Nominal GDP} / \text{GDP Deflator}) \times 100$$

$$\text{Real GDP 2012} = (225,000 / 125) \times 100$$

$$= 1,800 \times 100$$

$$= \$180,000.$$

Final answer: Real GDP 2012 = \$180,000.

Part A.ii — Inflation rate from 2011 to 2012

Solution

$$\text{Inflation} = (\text{Deflator}_2 - \text{Deflator}_1) / \text{Deflator}_1 \times 100$$

$$= (125 - 100) / 100 \times 100$$

$$= 25 / 100 \times 100$$

= 25%.

(GDP Deflator is used here since CPI isn't given. The base year is 2011 since deflator = 100 there.)

Final answer: Inflation rate = 25%.

Part B — Foreign demand for Fehran's electric vehicles

Solution (Direction + Explanation)

DIRECTION: Foreign importers will buy FEWER electric vehicles from Fehran in 2012 vs 2011.

EXPLANATION:

- Fehran's price level rose 25% (from Part A.ii).
- Trading partners' price level remained the same.
- Therefore, Fehran's electric vehicles became RELATIVELY MORE EXPENSIVE compared to alternatives in foreign markets.
- When Fehran's goods are relatively more expensive, foreign buyers substitute toward cheaper alternatives.
- Result: foreign demand for Fehran's electric vehicles DECREASES.

Final answer: FEWER. Reason: 25% inflation made EVs relatively more expensive abroad.

Part C — Standard of living

Solution

Standard of living = REAL GDP PER CAPITA.

2011: Real GDP = $(150,000 / 100) \times 100 = \$150,000$. Population = 100.

Real GDP per capita 2011 = $150,000 / 100 = \$1,500$.

2012: Real GDP = \$180,000 (from Part A.i). Population = 120.

Real GDP per capita 2012 = $180,000 / 120 = \$1,500$.

DIRECTION: Standard of living STAYED THE SAME between 2011 and 2012.

EXPLANATION: Real GDP grew (150K → 180K, +20%), but population grew at the same rate (100 → 120, +20%). Per-capita real GDP unchanged.

Final answer: Stayed the same.

Part D — Effect on national debt

Solution

Budget deficit = Government Outlays – Tax Revenues.

2011: Outlays \$25,000, Taxes \$25,000 → Deficit = \$0 (BALANCED budget).

2012: Outlays \$35,000, Taxes \$30,000 → Deficit = \$5,000.

Fehran went from a balanced budget in 2011 to a \$5,000 deficit in 2012.

DIRECTION: National debt INCREASED.

EXPLANATION: Whenever a country runs a deficit, it must BORROW to cover the gap, which adds to the national debt. The 2012 deficit of \$5,000 directly increases the debt by \$5,000.

Final answer: National debt rose by \$5,000.

12.6 — Worked Solution: Actual FRQ #2 (Zeetoland)

FRQ #2 setup (from your practice)

Zeetoland is in SHORT-RUN EQUILIBRIUM.

Expected inflation = 4%.

Actual unemployment = 5%.

NRU = 6%.

Equilibrium real interest rate = 3%.

Six parts: A (graph), B (self-correction), C (identify policy), D (3 effects), E (forex graph), F (exports reasoning).

Reading the setup — what gap is Zeetoland in?

Critical interpretation

Actual $U = 5\% < \text{NRU} = 6\%$. Unemployment is BELOW the natural rate.

→ The economy is OVERHEATING. Cyclical unemployment is NEGATIVE (sort of — really, U is below NRU because RGDP is above potential).

→ INFLATIONARY GAP. $\text{RGDP} > \text{potential GDP}$.

→ AD/SRAS intersect to the RIGHT of LRAS.

This is why the central bank in Part C is worried about inflation — Zeetoland is in an inflationary gap.

Part A — Graph (AD, SRAS, LRAS)

How to draw it

Title: 'AD/AS Model — Zeetoland.'

Y-axis: Price Level. X-axis: Real GDP.

Draw three curves:

- LRAS: vertical line at Y_F (full-employment output).
- SRAS: upward-sloping curve.
- AD: downward-sloping curve.

AD and SRAS must intersect to the RIGHT of LRAS (inflationary gap).

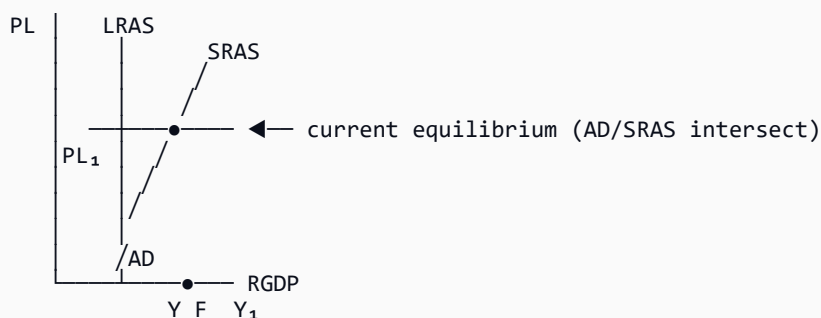
Mark the equilibrium point (intersection of AD and SRAS).

Drop dotted lines from equilibrium to:

- Y-axis: label PL_1 .
- X-axis: label Y_1 .

Drop a dotted line from the LRAS-X-axis intersection: label Y_F .

Y_1 should be visibly RIGHT of Y_F ($Y_1 > Y_F$).



Part B — Self-correction with no policy

Solution (full chain explanation)

Zeetoland is in an inflationary gap ($RGDP > \text{potential}$, $U < NRU$).

STEP 1: The labor market is TIGHT. Unemployment (5%) is below NRU (6%). Workers have bargaining power.

STEP 2: As a result, NOMINAL WAGES RISE.

STEP 3: Higher wages = higher input costs for firms.

STEP 4: Higher input costs cause SRAS to SHIFT LEFT.

STEP 5: This process continues until SRAS shifts far enough left that AD now intersects SRAS ON the LRAS.

STEP 6: At new long-run equilibrium: RGDP returns to Y_F (potential), unemployment returns to NRU (6%), and PRICE LEVEL is HIGHER than originally.

Final answer: SRAS shifts left until economy reaches LRAS at higher PL and Y_F .

Part C — Monetary policy action to fight inflation

Solution

Central bank wants CONTRACTIONARY monetary policy.

With ample reserves system, possible actions (any ONE is sufficient):

- RAISE the IORB (interest on reserve balances) — primary tool in ample reserves.
- SELL government bonds (open-market sale).
- RAISE the discount rate.
- RAISE the reserve requirement (rare but valid).

Final answer (recommended): RAISE the IORB rate. (Most common in modern Fed.)

Part D.i — Effect on bond prices in Zeetoland

Solution

Contractionary monetary action signals tighter policy → interest rates rise.

Bond prices and interest rates move INVERSELY.

Therefore: PRICE OF PREVIOUSLY ISSUED BONDS FALLS.

(Existing fixed-rate bonds become less attractive when new bonds are issued at higher rates.)

Final answer: Bond prices DECREASE.

Part D.ii — Effect on real output (with explanation)

Solution (Direction + Explanation)

DIRECTION: Real output DECREASES.

EXPLANATION:

1. Contractionary policy raises the nominal rate → real rate rises.
 2. Higher real interest rate makes borrowing more expensive.
 3. Investment spending (I) decreases. Durables-consumption decreases.
 4. AD shifts LEFT.
 5. With AD shifting left, the AD/SRAS intersection moves to a lower RGDP.
 6. Real output FALLS.
- Final answer: Real output decreases due to higher real rates → less I → AD left.

Part D.iii — Effect on net financial capital flows

Solution (Direction + Explanation)

DIRECTION: Net financial capital flows TO Zeetoland INCREASE (capital INFLOW).

EXPLANATION:

1. Contractionary monetary policy raises Zeetoland's real interest rate.
 2. Higher domestic real returns make Zeetoland's bonds and assets more attractive to foreign investors.
 3. Foreign investors purchase more Zeetoland assets → capital flows INTO Zeetoland.
 4. Therefore: net financial capital flow INTO Zeetoland INCREASES.
- Final answer: Capital INFLOW to Zeetoland increases.

Part E — Forex graph (zeet)**How to draw it**

Title: 'Foreign Exchange Market for the Zeet.'

Y-axis: Exchange rate (e.g., USD per zeet, or zeet's price in foreign currency).

X-axis: Quantity of zeet.

Draw two curves:

- D_{zeet} (downward-sloping). Original D_{zeet_1} .
- S_{zeet} (upward-sloping).

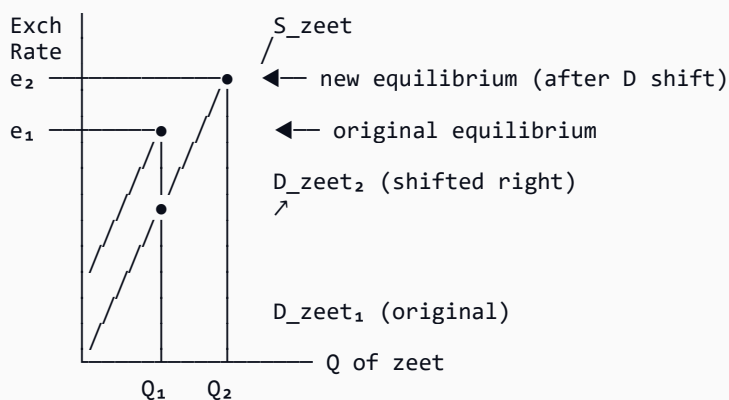
Mark original equilibrium (intersection) at exchange rate e_1 and quantity Q_1 .

After contractionary policy: foreign capital wants Zeetoland bonds → demand for zeet INCREASES.

→ D_{zeet} shifts RIGHT ($D_{\text{zeet}_1} \rightarrow D_{\text{zeet}_2}$). Show with arrow.

New equilibrium: higher exchange rate e_2 (zeet APPRECIATES) and higher quantity Q_2 .

Drop dotted lines for both equilibria, label all values.



Zeet APPRECIATES – exchange rate rises from e_1 to e_2 .

Part F — Effect on Zeetoland's exports**Solution (Direction + Explanation)**

DIRECTION: Zeetoland's EXPORTS DECREASE.

EXPLANATION:

1. From Part E, the zeet has APPRECIATED.
2. A stronger zeet means Zeetoland's goods are now MORE EXPENSIVE for foreign buyers (it takes more of their currency to buy a zeet, so it takes more of their currency to buy a Zeetoland-made good).
3. Foreign buyers substitute toward cheaper alternatives.
4. Demand for Zeetoland's exports falls.

Final answer: Exports decrease because the appreciated zeet makes Zeetoland's goods more expensive abroad.

12.7 — FRQ Common Patterns (with templates)

Pattern A: 'Recessionary gap' setup

Template

Original: AD/SRAS intersect LEFT of LRAS. RGDP < potential. U > NRU. Cyclical exists.

Expansionary fiscal: $\uparrow G$ or $\downarrow T \rightarrow AD$ right $\rightarrow PL \uparrow$, RGDP $\uparrow$, U $\downarrow$.

Real rate effect: gov borrows $\rightarrow DLF$ right $\rightarrow$ real rate $\uparrow$ (CROWDING OUT).

Expansionary monetary: Fed buys bonds $\rightarrow MS$ right $\rightarrow$ nominal rate $\downarrow \rightarrow$ real rate $\downarrow \rightarrow I \uparrow \rightarrow AD$ right.

Forex: lower US rates $\rightarrow$ capital outflow $\rightarrow S_USD$ right $\rightarrow$ dollar DEPRECIATES $\rightarrow X \uparrow$, M $\downarrow \rightarrow AD \uparrow$ more.

Pattern B: 'Inflationary gap' setup (Zeetoland)

Template

Original: AD/SRAS intersect RIGHT of LRAS. RGDP > potential. U < NRU.

Self-correction: tight labor $\rightarrow$ wages up $\rightarrow$ SRAS LEFT $\rightarrow$ return to LRAS at higher PL.

Contractionary monetary: Fed sells (or raises IORB) $\rightarrow MS$ left $\rightarrow$ nominal up $\rightarrow$ real up $\rightarrow I$ down $\rightarrow AD$ LEFT $\rightarrow PL \downarrow$, RGDP $\downarrow$.

Bond prices: FALL (rates rise).

Forex: higher rates $\rightarrow$ capital INFLOW $\rightarrow$ currency APPRECIATES $\rightarrow$ exports DECREASE.

Pattern C: Cost-push / negative supply shock

Template

SRAS shifts LEFT.

Result: $PL \uparrow$, $RGDP \downarrow$, $U \uparrow$ — STAGFLATION.

Phillips curve: SRPC SHIFTS RIGHT (worse trade-off).

Self-correction: high U eventually pushes wages down $\rightarrow$ SRAS shifts back right (slow).

Pattern D: Monetary policy traced through ALL markets

Template — full chain (the universal money chain)

(a) Money market: MS shifts (right if expansionary, left if contractionary) $\rightarrow$ nominal i moves.

(b) Loanable funds: real rate moves the same direction as nominal.

(c) AD/AS: I responds to real rate $\rightarrow$ AD shifts.

(d) Forex: capital flows respond to rate change $\rightarrow$ currency moves.

(e) Phillips: AD shift $\rightarrow$ MOVE along SRPC. Don't shift unless SRAS or expectations changed.

Show every shift on its respective graph with proper labels and dotted lines.

Pattern E: Long-run economic growth

Template

More capital, technology, or labor $\rightarrow$ LRAS shifts RIGHT.

PPC shifts outward (your Q13).

In the long run: $PL \downarrow$, $RGDP \uparrow$.

Growth requires SAVINGS $\rightarrow$ INVESTMENT in capital. Trace through loanable funds market.

Pattern F: Open economy questions

Template

Foreign income $\uparrow \rightarrow$ foreigners buy more US exports $\rightarrow D_{USD}$ right $\rightarrow$ dollar APPRECIATES.

Net exports $\uparrow \rightarrow$ AD right in US $\rightarrow RGDP \uparrow$, $PL \uparrow$, $U \downarrow$.

Current account improves.

12.8 — Time Management on FRQs

Recommended pacing

Reading period (10 min): outline answers for all 3 FRQs, sketch graphs lightly, identify which models you'll use.

Long FRQ (25 min): tackle parts in order. Skip a part if stuck — come back later. Each part is independent for grading.

Short FRQs (12 min each): same approach. Don't over-write.

Final 1 min: scan for unlabeled axes or curves. Add labels.

If you run out of time

Don't write full sentences. List your answers.

Sketch a quick graph with at least labeled axes and a labeled curve shifting in the right direction.

Partial points beat zero. The rubric rewards specific answers, not flowery prose.

Part 13 — Timed Practice

Pacing

30 MCQs in 35 minutes (matches AP pacing of ~70 sec each).

Then 1 long FRQ in 25 minutes.

Total: 60 minutes.

Use a timer. Write answers on a separate sheet. Don't peek at solutions until timer is up.

13.1 — 30 Multiple Choice Questions

1. If the MPC = 0.6, the spending multiplier is:
 - A) 1.4
 - B) 1.67
 - C) 2.5
 - D) 6.67
2. An expansionary monetary policy will most likely cause:
 - A) Real rate $\uparrow$, dollar appreciate
 - B) Real rate $\downarrow$, dollar depreciate
 - C) Real rate $\uparrow$, dollar depreciate
 - D) Real rate $\downarrow$, dollar appreciate
3. Crowding out occurs when:
 - A) Fed buys bonds and money supply rises
 - B) Government deficit raises real rates and reduces private investment
 - C) Imports fall due to currency depreciation
 - D) Inflation reduces consumer purchasing power
4. On the long-run Phillips curve, the unemployment rate is:
 - A) Zero
 - B) Cyclical only
 - C) Equal to NRU (frictional + structural)
 - D) Variable with inflation
5. Which is NOT in M1?

- A) Currency in circulation
 - B) Demand deposits
 - C) Savings deposits
 - D) Traveler's checks
6. Which scenario describes stagflation?
- A) PL $\uparrow$, RGDP $\uparrow$, U $\downarrow$
 - B) PL $\downarrow$, RGDP $\downarrow$, U $\uparrow$
 - C) PL $\uparrow$, RGDP $\downarrow$, U $\uparrow$
 - D) PL $\downarrow$, RGDP $\uparrow$, U $\downarrow$
7. If the dollar appreciates relative to other currencies:
- A) US X up, M down, AD right
 - B) US X down, M up, AD left
 - C) Both X and M up
 - D) No effect on trade
8. An increase in expected inflation will most likely:
- A) Shift SRAS right
 - B) Shift SRAS left
 - C) Shift LRAS right
 - D) Have no effect on AS
9. If RRR = 20%, the money multiplier is:
- A) 2
 - B) 4
 - C) 5
 - D) 8
10. Which is counted in GDP?
- A) Sale of a 2015 car
 - B) Stocks bought on the NYSE
 - C) Construction of a new home this year
 - D) Social Security check

11. Discouraged workers leaving the labor force:

- A) Increase the official unemployment rate
- B) Decrease the official rate while true joblessness is unchanged
- C) Have no effect on either
- D) Are counted as employed

12. An increase in the price level causes:

- A) AD to shift left
- B) Movement along AD (less RGDP demanded)
- C) AD to shift right
- D) SRAS to shift left

13. If both AD and SRAS shift right, what happens to RGDP and PL?

- A) Both rise
- B) RGDP rises, PL is indeterminate
- C) PL rises, RGDP indeterminate
- D) Both fall

14. An economy is at full employment. AD shifts RIGHT. In the LONG run:

- A) RGDP rises permanently
- B) PL rises, RGDP returns to potential
- C) U falls permanently
- D) Both PL and RGDP fall

15. Higher US real interest rates will most likely:

- A) Cause capital outflow and dollar depreciation
- B) Cause capital inflow and dollar appreciation
- C) Have no effect on capital flows
- D) Reduce US imports immediately

16. An economy has Nominal GDP = \$1,500B and Real GDP = \$1,200B. The GDP deflator is:

- A) 80
- B) 100
- C) 125
- D) 250

17. If $MV = PQ$ and M grows 6%, V is constant, Q grows 2%, inflation is approximately:

- A) 2%
- B) 4%
- C) 6%
- D) 8%

18. Buying a US Treasury bond on the secondary market:

- A) Counts as Investment in GDP
- B) Counts as Government spending
- C) Is NOT counted in GDP
- D) Counts as Consumption

19. Country X has a current account deficit of \$200B. Its financial account must be:

- A) Deficit of \$200B
- B) Surplus of \$200B
- C) Balanced
- D) Cannot be determined

20. A bank has \$20,000 in deposits, $RRR = 10\%$. Required reserves are:

- A) \$200
- B) \$2,000
- C) \$5,000
- D) \$20,000

21. If the same bank has total reserves of \$5,000, excess reserves are:

- A) \$0
- B) \$1,000
- C) \$3,000
- D) \$5,000

22. The maximum money supply increase from those excess reserves is:

- A) \$3,000
- B) \$10,000
- C) \$30,000

D) \$50,000

23. An increase in foreign income is most likely to:

- A) Decrease US exports
- B) Increase US net exports and shift AD right
- C) Have no effect on US AD
- D) Cause USD to depreciate

24. An economy with cyclical unemployment is in:

- A) Inflationary gap
- B) Long-run equilibrium
- C) Recessionary gap
- D) Stagflation

25. Long-run economic growth requires:

- A) Lower interest rates only
- B) Capital formation, technology, or labor force expansion
- C) Higher consumer spending
- D) Lower government deficits

26. If government spending rises \$200B and taxes rise \$200B, the change in GDP is:

- A) \$0
- B) \$200B
- C) \$400B
- D) Depends on MPC

27. The Fed sells government bonds. The most direct effect is:

- A) MS rises and rates fall
- B) MS falls and rates rise
- C) Reserve requirement rises
- D) Discount rate rises

28. Real interest rate is the:

- A) Nominal rate plus inflation
- B) Nominal rate minus inflation

- C) Inflation rate alone
- D) Same as the nominal rate

29. Government tax cut of \$300B with $MPC = 0.8$ raises GDP by:

- A) \$300B
- B) \$600B
- C) \$1,200B
- D) \$1,500B

30. A negative supply shock will most likely cause:

- A) Both PL and RGDP to fall
- B) PL to rise and RGDP to rise
- C) PL to rise and RGDP to fall (stagflation)
- D) PL to fall and RGDP to rise

13.2 — Long FRQ (25 minutes) — Mirrors Your Practice Style

Assume the economy of Tirvania is currently operating in a recessionary gap. The Tirvi (Tirvania's currency) is freely traded on the foreign exchange market.

1. Draw a correctly labeled AD/AS graph showing the recessionary gap. Label current price level PL_i , current real GDP Y_i , and full-employment output Y_F .
2. Suppose the central bank takes appropriate monetary policy to close the gap. (i) What open-market operation is appropriate? (ii) Show the effect on a correctly labeled money market graph. (iii) What happens to the nominal interest rate?
3. Using a correctly labeled loanable funds market graph, show the effect of the central bank's policy on the real interest rate.
4. How will the change in real interest rate affect investment? Explain.
5. Using a correctly labeled forex market graph for the Tirvi, show the effect of the central bank's policy on the value of the Tirvi. Will the Tirvi appreciate, depreciate, or stay the same?
6. How does the change in the Tirvi's value affect Tirvania's net exports? Explain.
7. Combining all effects, will the policy succeed in closing the recessionary gap? Show the new equilibrium on your AD/AS graph from part (a).
8. In the long run, will the policy affect the natural rate of unemployment? Explain.

13.3 — MCQ Answer Key

1. C — $Mult = 1/(1-0.6) = 1/0.4 = 2.5$.

2. **B** — Fed adds liquidity → real rate ↓. Lower US rates → capital outflow → dollar depreciates.
3. **B** — Crowding out is in loanable funds: gov borrowing → DLF right → real rate up → I down.
4. **C** — LRPC is vertical at NRU. No long-run trade-off.
5. **C** — Savings deposits are in M2, not M1.
6. **C** — Stagflation = SRAS shifts left → PL up, RGDP down, U up.
7. **B** — Strong dollar makes US goods expensive abroad (X down) and foreign goods cheaper (M up).
8. **B** — Workers anticipate higher prices → demand higher wages → input costs rise → SRAS LEFT.
9. **C** — Money mult = $1/RRR = 1/0.20 = 5$.
10. **C** — New construction counts as Investment (I).
11. **B** — They leave LF, so numerator and denominator both fall — denominator falls more proportionally → rate falls.
12. **B** — PL changes cause MOVEMENT along AD (the three effects).
13. **B** — Same direction shifts: quantity (RGDP) determined, price (PL) ambiguous.
14. **B** — LR self-correction: tight labor → wages up → SRAS left → return to LRAS at higher PL.
15. **B** — Foreign investors seek higher US returns → capital inflow → demand for USD rises → dollar appreciates.
16. **C** — Deflator = $(\text{Nominal}/\text{Real}) \times 100 = (1,500/1,200) \times 100 = 125$.
17. **B** — $\% \Delta P \approx \% \Delta M - \% \Delta Q = 6 - 2 = 4\%$.
18. **C** — Financial transactions don't count in GDP — no production occurred.
19. **B** — $CA + FA = 0$. Deficit in one ⇒ surplus in the other.
20. **B** — Required = Deposits $\times$ RRR = $20,000 \times 0.10 = \$2,000$.
21. **C** — Excess = Total – Required = $5,000 - 2,000 = \$3,000$.
22. **C** — Money mult = $1/0.10 = 10$. $\$3,000 \times 10 = \$30,000$.
23. **B** — Foreign income up → foreigners buy more US exports → X up → AD right. Also: D_{USD} up, dollar appreciates.
24. **C** — Cyclical unemployment = recessionary gap. $RGDP < \text{potential}$, $U > NRU$.
25. **B** — Growth = LRAS shifts right = more resources, better tech, more capital, better human capital.
26. **B** — Balanced budget multiplier = 1. $\Delta GDP = \Delta \text{Spending} = \$200B$.
27. **B** — Selling bonds → bond prices fall → rates rise. Reserves drained from system → MS falls.
28. **B** — Real $\approx$ Nominal – Inflation (Fisher equation).
29. **C** — Tax mult = $-0.8/0.2 = -4$. $\Delta GDP = 300 \times 4 = \$1,200B$.
30. **C** — SRAS shifts left → stagflation: PL up, RGDP down, U up.

13.4 — Long FRQ Solution

(a) AD/AS graph: AD intersects SRAS LEFT of LRAS. $PL_1 < \text{long-run PL}$. $Y_1 < Y_F$. Dotted lines from intersection to PL_1 and Y_1 . Dotted line from LRAS to Y_F .

(b)(i) Central bank should BUY government bonds (open-market purchase).

(b)(ii) Money market: MS shifts RIGHT (MS_1 to MS_2). Nominal rate falls from i_1 to i_2 .

(b)(iii) Nominal interest rate FALLS.

(c) Loanable funds: SLF shifts RIGHT. Real interest rate falls from r_1 to r_2 .

(d) Lower real interest rate REDUCES the cost of borrowing for firms. Therefore, business INVESTMENT INCREASES — firms borrow to fund capital projects that are now profitable at the lower rate.

(e) Forex (Tirvi): lower domestic real rates make Tirvania bonds less attractive to foreign investors → capital outflow → S_{Tirvi} shifts right (or D_{Tirvi} shifts left) → TIRVI DEPRECIATES.

(f) Weaker Tirvi makes Tirvania's goods CHEAPER for foreigners (X up) and foreign goods MORE EXPENSIVE for Tirvanians (M down). Net exports INCREASE.

(g) Combined: investment ↑ AND net exports ↑ → AD shifts RIGHT. New equilibrium at higher PL and higher RGDP. If calibrated correctly, RGDP reaches Y_F , closing the gap.

(h) NO. The natural rate of unemployment is determined by frictional + structural unemployment, which depend on labor market institutions, demographics, and technology — NOT on monetary policy. Monetary policy can affect cyclical unemployment but not the natural rate.

Part 14 — One-Page Cheat Sheet

Print this. Carry it. Review it on the way to the exam if you choose to do exam-morning prep.

Formulas

$$\begin{aligned} \% \text{ change} &= (\text{New} - \text{Old}) / \text{Old} \times 100 \\ \text{GDP} &= C + I + G + (X - M) \\ \text{Real GDP} &= \text{Nominal GDP} / \text{GDP Deflator} \times 100 \\ \text{Real GDP per capita} &= \text{Real GDP} / \text{Population} \\ \text{Real Wage} &= \text{Nominal Wage} / \text{CPI} \times 100 \\ \text{Real } i &\approx \text{Nominal } i - \text{Inflation} \\ \text{Inflation} &= (\text{CPI}_2 - \text{CPI}_1) / \text{CPI}_1 \times 100 \\ \text{GDP Deflator} &= (\text{Nominal/Real}) \times 100 \\ \\ \text{Labor Force} &= \text{Employed} + \text{Unemployed (actively seeking)} \\ \text{U Rate} &= \text{Unemployed} / \text{LF} \times 100 \\ \text{LFPR} &= \text{LF} / \text{Working-age Pop} \times 100 \\ \text{NRU} &= \text{Frictional} + \text{Structural} \\ \\ \text{MPC} + \text{MPS} &= 1 \\ \text{Spending Multiplier} &= 1/\text{MPS} = 1/(1-\text{MPC}) \\ \text{Tax Multiplier} &= -\text{MPC}/\text{MPS} \\ \text{Balanced Budget Multiplier} &= 1 \text{ (ALWAYS)} \\ \Delta \text{GDP} &= \Delta \text{Spending} \times \text{Mult} \quad \text{OR} \quad \Delta \text{Taxes} \times \text{Tax Mult} \\ \\ \text{Required Reserves} &= \text{Deposits} \times \text{RRR} \\ \text{Money Multiplier} &= 1/\text{RRR} \\ \text{Max } \Delta \text{MS} &= \text{Excess Reserves} \times \text{Money Mult} \\ \\ \text{MV} = \text{PQ} &\rightarrow \% \Delta \text{M} \approx \% \Delta \text{P} + \% \Delta \text{Q} \text{ (V constant)} \\ \text{Current Account} + \text{Financial Account} &= 0 \end{aligned}$$

Common multiplier values

MPC	Spending Mult	Tax Mult
0.50	2	-1
0.60	2.5	-1.5
0.75	4	-3
0.80	5	-4
0.90	10	-9

RRR	Money Mult
10%	10
20%	5
25%	4
50%	2

Four shift outcomes (every market)

D right: $\uparrow P, \uparrow Q$	BOTH same direction:
D left: $\downarrow P, \downarrow Q$	$\rightarrow Q$ determined, P INDETERMINATE
S right: $\downarrow P, \uparrow Q$	
S left: $\uparrow P, \downarrow Q$	BOTH opposite directions:
	$\rightarrow P$ determined, Q INDETERMINATE

Effect lookup — every shock

SHOCK	PL	RGDP	U	Real i	Currency
Expansionary fiscal	$\uparrow$	$\uparrow$	$\downarrow$	$\uparrow$	Appreciate
Contractionary fiscal	$\downarrow$	$\downarrow$	$\uparrow$	$\downarrow$	Depreciate
Expansionary monetary	$\uparrow$	$\uparrow$	$\downarrow$	$\downarrow$	Depreciate
Contractionary monetary	$\downarrow$	$\downarrow$	$\uparrow$	$\uparrow$	Appreciate
Positive AD shock	$\uparrow$	$\uparrow$	$\downarrow$	$\uparrow$	Appreciate
Negative AD shock (recession)	$\downarrow$	$\downarrow$	$\uparrow$	$\downarrow$	Depreciate
Positive supply shock	$\downarrow$	$\uparrow$	$\downarrow$	$\downarrow$	Depreciate
Negative supply (stagflation)	$\uparrow$	$\downarrow$	$\uparrow$	$\uparrow$	Mixed
Long-run growth (LRAS right)	$\downarrow LR$	$\uparrow$	$\downarrow$	$?$	$?$
Foreign income $\uparrow$	$\uparrow$	$\uparrow$	$\downarrow$	$?$	Appreciate

Compound policy table

COMBO	AD	Real Rate
Both expansionary	$\uparrow\uparrow$	INDETERMINATE
Both contractionary	$\downarrow\downarrow$	INDETERMINATE
Expansionary + Contractionary	INDETERM	Definite (depends on which)
Contractionary + Expansionary	INDETERM	Definite (depends on which)

The 12 chains — quick reference

- **Expansionary fiscal:** AD right $\rightarrow$ DLF right $\rightarrow$ real rate up (CROWDING OUT) $\rightarrow$ dollar appreciates $\rightarrow X$ down
- **Contractionary fiscal:** AD left, real rate down, dollar depreciates
- **Expansionary monetary:** MS right $\rightarrow$ nominal i down $\rightarrow$ real i down $\rightarrow I$ up $\rightarrow$ AD right $\rightarrow$ dollar depreciates $\rightarrow X$ up
- **Contractionary monetary:** MS left $\rightarrow$ real i up $\rightarrow I$ down $\rightarrow$ AD left $\rightarrow$ bond prices DOWN $\rightarrow$ capital INFLOW $\rightarrow$ dollar APPRECIATES $\rightarrow X$ down
- **Positive supply shock:** SRAS right $\rightarrow$ PL down, RGDP up, U down. SRPC LEFT.
- **Negative supply shock:** SRAS left $\rightarrow$ STAGFLATION. SRPC RIGHT.
- **LR growth:** LRAS right $\rightarrow$ PPC outward. Long-run PL falls, RGDP rises.

- **Foreign income up:** X up $\rightarrow D_{USD}$ right $\rightarrow$ dollar appreciates $\rightarrow$ AD right.

Critical traps

- **Money market vs Loanable funds:** MM = nominal rate, Fed-driven. LF = real rate. Crowding out is in LF.
- **Movement vs shift:** Own-price change = movement. Determinant change = shift.
- **Multiply vs divide:** Find ΔGDP from $\Delta Spending$ $\rightarrow$ MULTIPLY. Find $\Delta Spending$ to close gap $\rightarrow$ DIVIDE.
- **Real vs nominal:** Investment driven by REAL rate. Nominal up but inflation up more $\rightarrow$ real fell $\rightarrow I$ up.
- **Long-run neutrality of money:** Monetary has NO long-run effect on real GDP.
- **Discouraged workers:** NOT in labor force. Excluded $\rightarrow$ official rate UNDERSTATES true unemployment.
- **CA + FA = 0:** Trade deficit $\Rightarrow$ capital inflow.
- **Stagflation signature:** PL up AND RGDP down $\rightarrow$ SRAS shifted LEFT.
- **Per capita:** Standard of living = REAL GDP / population. RGDP growth alone doesn't mean rising living standards.
- **PPC + LRAS:** ALWAYS shift together (Q13). Tech up $\rightarrow$ both outward.
- **Bond prices and rates:** Always INVERSE. Contractionary monetary $\rightarrow$ bond prices FALL.
- **Inventories:** ARE counted in GDP (as Investment).

FRQ checklist (every graph)

- 1. Title at top
- 2. Both axes labeled
- 3. All curves labeled (D, S, AD, SRAS, LRAS, MD, MS, DLF, SLF, etc.)
- 4. Equilibrium dotted lines to BOTH axes, labeled (P^* , Q^* , i^* , r^* , e^*)
- 5. If shifted: original AND new curves labeled
- 6. Direction-of-change arrows visible

Exam strategy

1. MCQ: two-pass. First pass = easy questions in <60s. Second = harder ones. Final = guess any blanks (no penalty).
2. FRQ: use 10-min reading period to outline. Long FRQ = 25 min. Short FRQs = 12 min each.
3. Read every question twice. Answer EXACTLY what's asked.
4. Label every graph. Always.
5. Never leave anything blank.
6. If finished early, RECHECK GRAPHS for missing labels.

Exam-day mantra

Identify which market the shock hits FIRST.

Trace the chain step by step.

Read PL, RGDP, U off the AD/AS hub.

Then trace into Phillips curve and forex if needed.

Every macro question is just a chain. You know the chains. Trust the work.